Use trig identities to solve the trig equations for $0 \leq x \leq 2\pi$. Find exact values.

9. $\sin^2 x(\csc x + 1) = 0$

10. $\frac{\cos^2 x}{\cot^2 x} = 1$

Answers to 3.12A CA #1

1. $\tan x$	2. $\cos x$	3. $\csc^2 x$
4. $\tan \theta$	5. $\tan^3 x$	6. $\sec^2 \theta$
7. $x = 0, \frac{\pi}{4}, \pi, \frac{5\pi}{4}, 2\pi$	8. $x = \frac{\pi}{3}, \frac{5\pi}{3}$	
9. $x = \frac{3\pi}{2}$ ($0, \pi$, and 2π are extraneous solutions)	10. $x = \frac{\pi}{2}, \frac{3\pi}{2}$	

Sum/Difference Identities

$\sin(\alpha + \beta) =$

$\sin(\alpha - \beta) =$

$\cos(\alpha + \beta) =$

$\cos(\alpha - \beta) =$

Find the exact value of the sum or difference.

$\sin\left(\frac{\pi}{4} + \frac{5\pi}{3}\right)$

$\cos\left(\frac{7\pi}{4} - \frac{2\pi}{3}\right)$

Simplify the following.

$\sin\left(x - \frac{\pi}{6}\right)$

Write your questions and thoughts here!

Double Angle Identities

$\sin(2\alpha) =$

$\cos(2\alpha) =$

Use trig identities to solve the trig equations for $0 \leq x \leq 2\pi$. Find exact values.

$\sin(2x) = \cos x$

$3\cos(2x) = 2\cos^2 x$

3.12B Practice

Find the exact value of the sum or difference.

1. $\sin\left(\frac{5\pi}{4}+\frac{2\pi}{3}\right)$

2. $\cos\left(\frac{3\pi}{4}+\frac{\pi}{3}\right)$

3. $\cos\left(\frac{5\pi}{4}-\frac{\pi}{6}\right)$

4. $\sin\left(\frac{3\pi}{2}-\frac{4\pi}{3}\right)$

5. $\cos\left(\frac{\pi}{2}-\frac{4\pi}{3}\right)$

6. $\sin\left(\frac{11\pi}{6}+\pi\right)$

Simplify the following.

7. $\sin\left(x - \frac{5\pi}{3}\right)$

8. $\cos\left(\frac{3\pi}{4} + \theta\right)$

9. $4\sin(2x)$

Use trig identities to solve the trig equations for $0 \leq x \leq 2\pi$. Find exact values.

10. $\sin(2x) + \sin x = 0$

11. $\sin(2x)\sec x = 0$

12. $\cos(2x) = \cos^2 x$

13. $\cos(2x) + \sin^2 x = 0$

14. The function f is given by $f(\theta) = \cos(2\theta)$. Which of the following expressions is equivalent to $f(\theta)$?

(A) $2\cos\theta$

(B) $1 - 2\sin^2\theta$

(C) $1 - 2\cos^2\theta$

(D) $2\sin\theta\cos\theta$

15. The function g is defined by $g(x) = \sin\left(x + \frac{\pi}{3}\right)$. The solutions to which of the following equations on the interval $0 \le x \le 2\pi$ are the solutions to $g(x) = 1$ on the interval $0 \le x \le 2\pi$?

(A) $\sin x + \sqrt{3}\cos x = 2$

(B) $\sin x - \sqrt{3}\cos x = 2$

(C) $\sqrt{3}\sin x + \cos x = 2$

(D) $\sqrt{3}\sin x - \cos x = 2$

16. The function f is given by $f(\theta) = \cos\left(\theta + \frac{3\pi}{2}\right)$. What is the value of $f\left(\frac{\pi}{3}\right)$?

(A) $\frac{\sqrt{3}}{2}$

(B) $\frac{1}{2}$

(C) $-\frac{1}{2}$

(D) $-\frac{\sqrt{3}}{2}$

17. Given $\cos(2x) = \frac{3}{5}$ and $\frac{\pi}{2} \le x \le \pi$, what is the exact value of $\sin(x)$?

(A) $\frac{2\sqrt{5}}{5}$

(B) $\frac{\sqrt{5}}{5}$

(C) $-\frac{\sqrt{5}}{5}$

(D) $-\frac{2\sqrt{5}}{5}$

3.12B Equivalent Representations of Trig Functions

AP Precalculus Name: _____ **CA #1**

Find the exact value of the sum or difference.

1. $\sin\left(\frac{3\pi}{4}-\frac{5\pi}{3}\right)$

2. $\cos\left(\frac{\pi}{2}+\frac{4\pi}{3}\right)$

3. $\cos\left(\pi-\frac{7\pi}{6}\right)$

4. $\sin\left(\frac{7\pi}{4}-\frac{\pi}{6}\right)$

Simplify the following.

5. $\sin\left(x-\frac{4\pi}{3}\right)$

6. $\cos\left(\frac{3\pi}{2}+\theta\right)$

7. $\cos(x-\pi)$

Use trig identities to solve the trig equations for $0 \leq x \leq 2\pi$. Find exact values.

8. $\cos(2x) = -\sin^2 x$

9. $\cos(2x)\sec^2 x = 0$

10. $\sin(2x) - \cos x = 0$

Answers to 3.12B CA #1

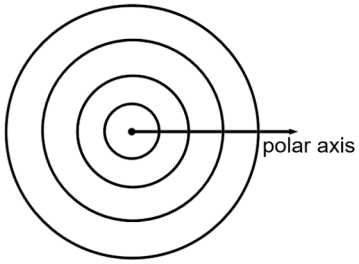
1. $\frac{\sqrt{2}-\sqrt{6}}{4}$	2. $\frac{\sqrt{3}}{2}$	3. $\frac{\sqrt{3}}{2}$
4. $\frac{-\sqrt{6}-\sqrt{2}}{4}$	5. $-\frac{1}{2}\sin x + \frac{\sqrt{3}}{2}\cos x$	6. $\sin\theta$
7. $-\cos x$	8. $x = \frac{\pi}{2}, \frac{3\pi}{2}$	
9. $x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$	10. $x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$	

3.13 Trigonometry and Polar Coordinates

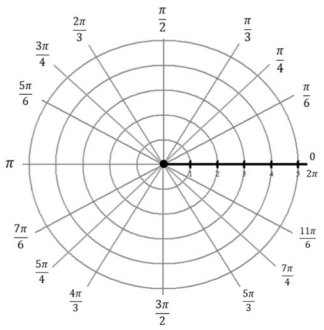
3.13 Notes

Polar Coordinates (r, θ)

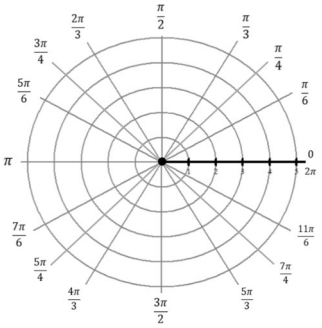
Pole:



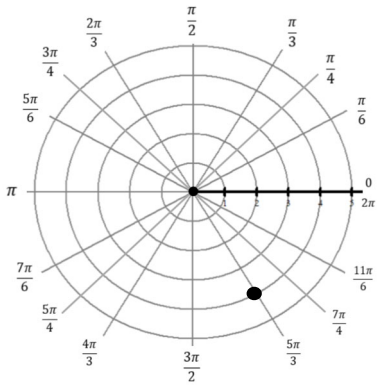
$\left(2, \frac{\pi}{6}\right)$



$\left(-3, \frac{7\pi}{6}\right)$



Name the point 4 different ways.



Convert polar to rectangular coordinates.

Formula:

$x =$

$y =$

$\left(6, \frac{\pi}{3}\right)$

$\left(4, \frac{5\pi}{4}\right)$

Convert rectangular to polar coordinates where $0 \leq \theta \leq 2\pi$.

Formula:

$r =$

$\theta =$

$(-3, 5)$

$(2, -6)$

Write your questions and thoughts here!

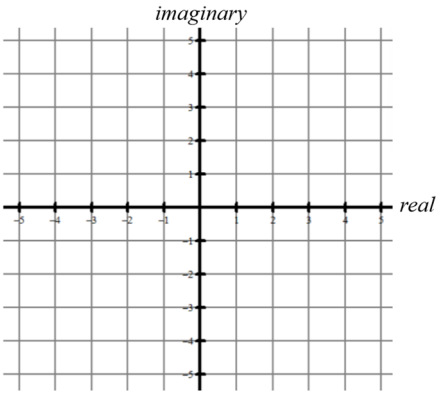
Complex Numbers $a + bi$

Absolute Value of Complex Number

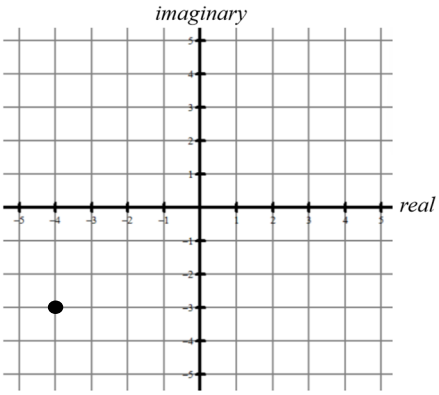
$z = a + bi$

$|z| = \sqrt{a^2 + b^2}$

$z = 3 + 4i$



Name the point.



Convert Rectangular Complex Numbers to Polar Form.

Formula:

$r =$

$\theta =$

$-3 + i\sqrt{7}$

$2 - i$

Convert Polar Complex Numbers to Rectangular Form.

Formula:

$x =$

$y =$

$3\cos\left(\frac{\pi}{3}\right) + 3i\sin\left(\frac{\pi}{3}\right)$

$4\left[\cos\left(\frac{3\pi}{2}\right) + i\sin\left(\frac{3\pi}{2}\right)\right]$

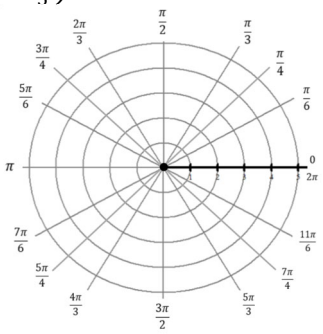
3.13 Trigonometry and Polar Coordinates

AP Precalculus

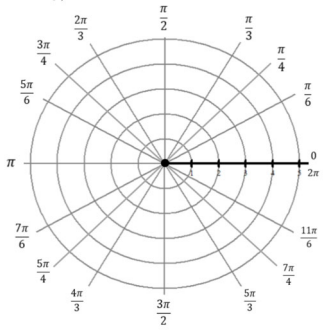
3.13 Practice

Plot the following polar points.

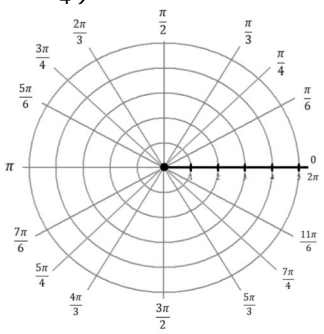
1. $\left(3, \frac{2\pi}{3}\right)$



2. $\left(-2, \frac{\pi}{6}\right)$

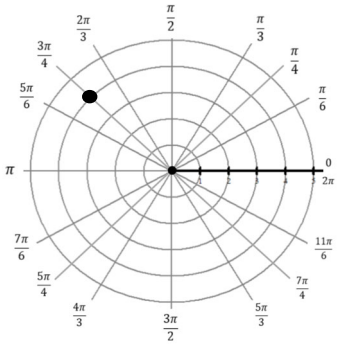


3. $\left(4, -\frac{3\pi}{4}\right)$

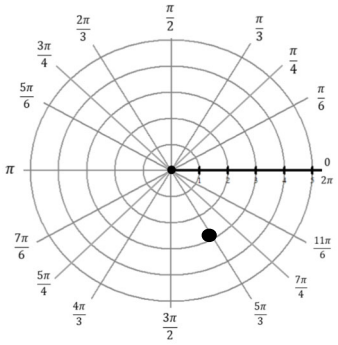


Name the polar point four different ways.

4.



5.



Convert the polar coordinates to rectangular coordinates.

6. $\left(4, \frac{2\pi}{3}\right)$

7. $\left(-2, \frac{\pi}{4}\right)$

8. $\left(5, \frac{5\pi}{6}\right)$

Convert the rectangular coordinates to polar coordinates where $0 \leq \theta \leq 2\pi$.

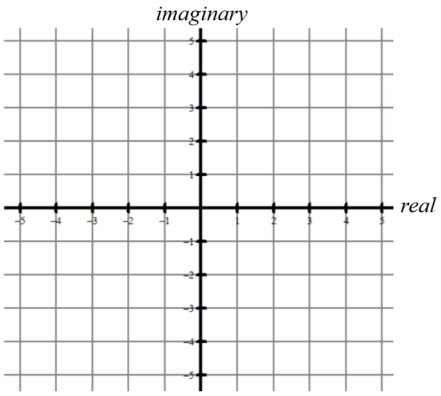
9. $(2, 4)$

10. $(-3, 4)$

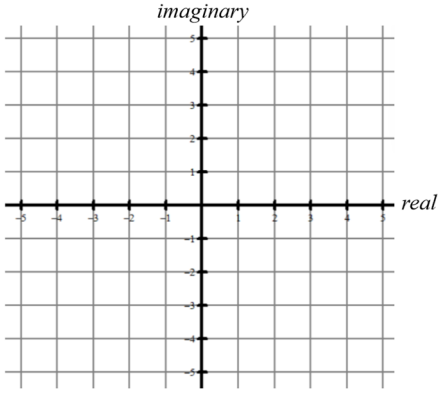
11. $(2, -1)$

Plot the following points on the complex plane.

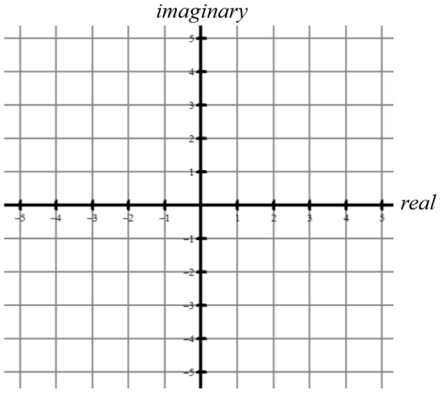
12. $3 - 4i$



13. $-4 + i$



14. $1 + 2i$



Convert rectangular complex numbers to polar form.

15. $3 + 4i$

16. $2 - 5i$

17. $-3 - 3i$

Convert polar complex numbers to rectangular form.

18. $10\left[\cos\left(\frac{3\pi}{4}\right) + i\sin\left(\frac{3\pi}{4}\right)\right]$

19. $4\left[\cos\left(\frac{3\pi}{2}\right) + i\sin\left(\frac{3\pi}{2}\right)\right]$

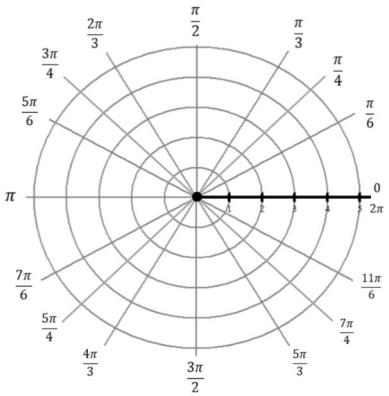
20. $3\cos\left(\frac{5\pi}{3}\right) + 3i\sin\left(\frac{5\pi}{3}\right)$

Fill in the table with approximate decimal values using a calculator. Plot the points.

21. $r = 4\cos\theta$

$\boldsymbol{\theta}$	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\boldsymbol{r}$					

$\boldsymbol{\theta}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π
$\boldsymbol{r}$				



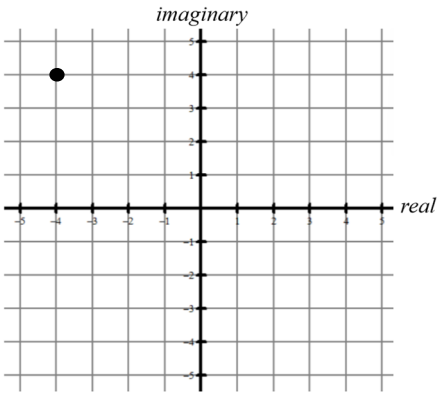
22. A complex number has the rectangular coordinates $(1, -\sqrt{3})$ Which of the following is one way to Express the complex number using its polar coordinates (r, θ) ?

(A) $\left(2\cos\left(-\frac{\pi}{6}\right)\right) + i\left(2\sin\left(-\frac{\pi}{6}\right)\right)$

(B) $\left(2\cos\left(-\frac{\pi}{3}\right)\right) + i\left(2\sin\left(-\frac{\pi}{3}\right)\right)$

(C) $\left(2\cos\left(\frac{\pi}{6}\right)\right) + i\left(2\sin\left(\frac{\pi}{6}\right)\right)$

(D) $\left(2\cos\left(\frac{\pi}{3}\right)\right) + i\left(2\sin\left(\frac{\pi}{3}\right)\right)$



23. The graph above indicates a point in the complex plane. The complex number has polar coordinates (r, θ). Which of the following completes the expression for the point in polar form $(4\sqrt{2}\cos\theta) + i(4\sqrt{2}\sin\theta)$?

(A) $\theta = \frac{\pi}{4}$

(B) $\theta = \frac{3\pi}{4}$

(C) $\theta = \frac{5\pi}{4}$

(D) $\theta = \frac{7\pi}{4}$

24. The point A has polar coordinates $\left(4, \frac{7\pi}{6}\right)$. Which of the following also gives the location of point A in polar coordinates?

(A) $\left(4, -\frac{11\pi}{6}\right)$

(B) $\left(4, -\frac{5\pi}{6}\right)$

(C) $\left(-4, -\frac{\pi}{6}\right)$

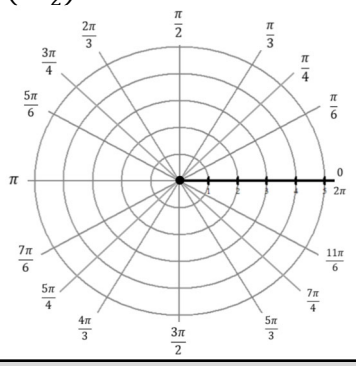
(D) $\left(-4, -\frac{5\pi}{6}\right)$

3.13 Trigonometric and Polar Coordinates

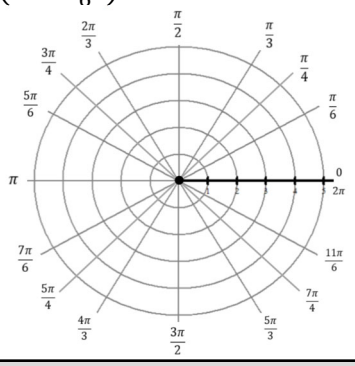
AP Precalculus Name: _____ **CA #1**

Plot the following polar points.

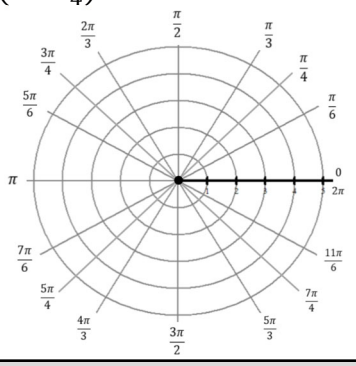
1. $\left(3, \frac{\pi}{2}\right)$



2. $\left(-4, \frac{11\pi}{6}\right)$



3. $\left(2, -\frac{\pi}{4}\right)$



Convert the polar coordinates to rectangular coordinates.

4. $\left(6, \frac{\pi}{3}\right)$

5. $\left(2, \frac{5\pi}{4}\right)$

6. $\left(-5, \frac{5\pi}{6}\right)$

Convert the rectangular coordinates to polar coordinates where $0 \leq \theta \leq 2\pi$.

7. $(-2, 4)$

8. $(-3, -4)$

9. $(4, -5)$

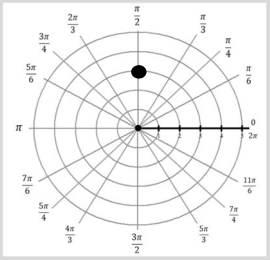
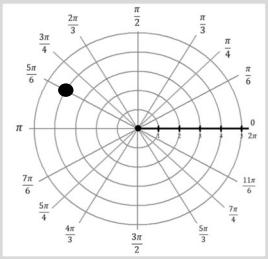
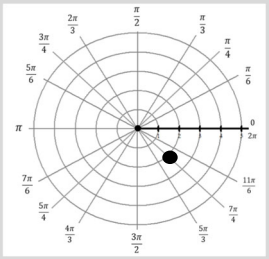
Convert rectangular complex numbers to polar form.

10. $3-4i$	11. $-3+5i$	12. $-3-2i$

Convert polar complex numbers to rectangular form.

13. $4\left[\cos\left(\frac{3\pi}{4}\right)+i\sin\left(\frac{3\pi}{4}\right)\right]$	14. $6\left[\cos\left(\frac{\pi}{6}\right)+i\sin\left(\frac{\pi}{6}\right)\right]$	15. $5\cos\left(\frac{4\pi}{3}\right)+5i\sin\left(\frac{4\pi}{3}\right)$

Answers to 3.13 CA #1

1. 	2. 	3. 
4. $(3, 3\sqrt{3})$	5. $(-\sqrt{2}, -\sqrt{2})$	6. $\left(\frac{5\sqrt{3}}{2}, -\frac{5}{2}\right)$
7. $(2\sqrt{5}, 2.034)$	8. $(5, 4.068)$	9. $(\sqrt{41}, 5.387)$
10. $5[\cos(-0.927)+i\sin(-0.927)]$	11. $\sqrt{34}[\cos(2.111)+i\sin(2.111)]$	12. $\sqrt{13}[\cos(3.729)+i\sin(3.729)]$
13. $-2\sqrt{2}+2i\sqrt{2}$	14. $3\sqrt{3}+3i$	15. $-\frac{5}{2}-\frac{5i\sqrt{3}}{2}$

AP Precalc

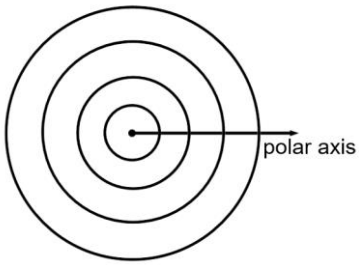
Write your questions and thoughts here!

3.14A Polar Function Graphs

3.14A Notes

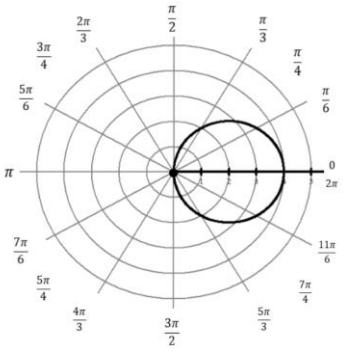
Lines

$\theta = \frac{\pi}{3}$



Circles

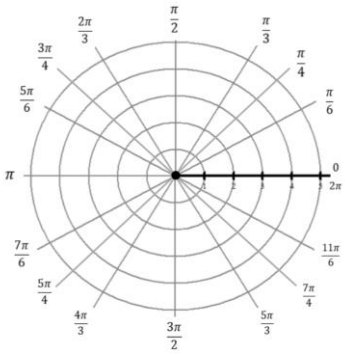
$r = 4\cos\theta$



Cycle:

Positive	Negative

$r = 4\sin\theta$



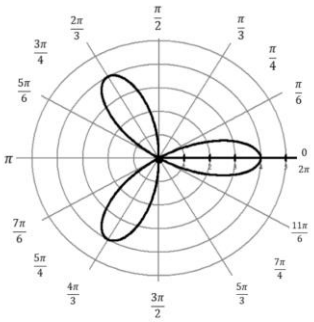
Cycle:

Positive	Negative

Roses

Odd *n* Cosine

$r = 4\cos(3\theta)$

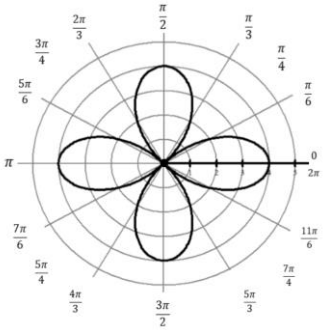


of petals:

Cycle:

Even *n* Cosine

$r = 4\cos(2\theta)$

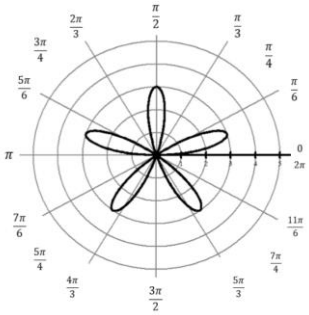


of petals:

Cycle:

Odd *n* Sine

$r = 3\sin(5\theta)$

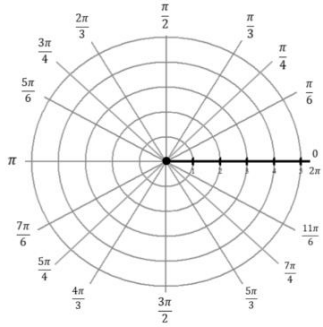


of petals:

Cycle:

Even *n* Sine

$r = 5\sin(4\theta)$



of petals:

Cycle:

Write your questions and thoughts here!

Describe the polar function.

$r = 2\cos(7\theta)$

Type:

Line Circle Rose

Opens: Petals:

Max distance from pole:

Cycle:

$r = 9\sin(\theta)$

Type:

Line Circle Rose

Opens: Petals:

Max distance from pole:

Cycle:

$r = 8\sin(6\theta)$

Type:

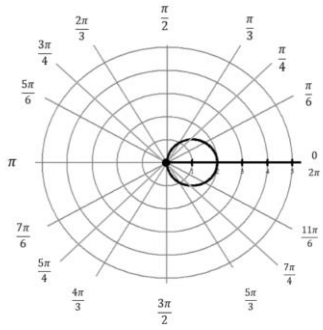
Line Circle Rose

Opens: Petals:

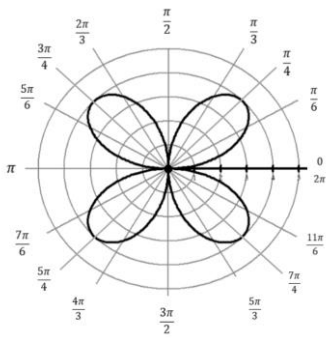
Max distance from pole:

Cycle:

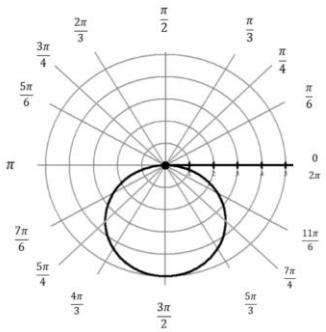
Write the equation of the following polar functions. Answer the questions.



Equation:



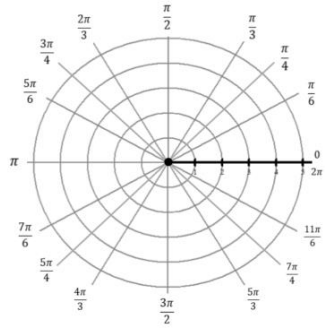
Equation:



Equation:

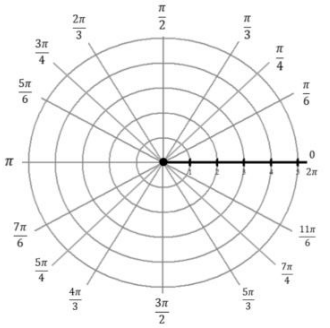
Sketch a graph. Find the endpoints of the restricted domain and highlight on the function.

$r = 3$



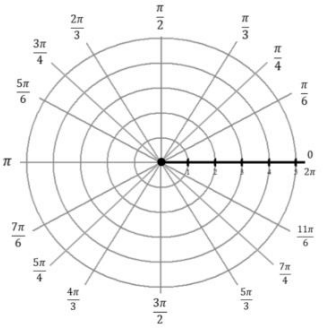
Endpoints of $\frac{\pi}{6} \le \theta \le \pi$

$r = 4\cos(3\theta)$



Endpoints of $\frac{\pi}{6} \le \theta \le \frac{\pi}{3}$

$r = 5\sin(\theta)$



Endpoints of $\frac{\pi}{2} \le \theta \le \pi$

3.14A Polar Function Graphs

AP Precalculus

3.14A Practice

Describe the equation of the polar function. Fill in the table.

1. $r = 6\cos(5\theta)$

Type:

Line Circle Rose

Opens: Petals:

Max distance from pole:

Cycle:

θ	r
$\frac{\pi}{6}$	
π	

2. $r = 4\sin(\theta)$

Type:

Line Circle Rose

Opens: Petals:

Max distance from pole:

Cycle:

θ	r
$\frac{\pi}{3}$	
$\frac{\pi}{2}$	

3. $r = 7$

Type:

Line Circle Rose

Center: Petals:

Max distance from pole:

Cycle:

θ	r
$\frac{\pi}{4}$	
$\frac{3\pi}{2}$	

4. $r = \cos(6\theta)$

Type:

Line Circle Rose

Opens: Petals:

Max distance from pole:

Cycle:

θ	r
$\frac{\pi}{6}$	
$\frac{\pi}{2}$	

5. $\theta = \frac{2\pi}{3}$

Type:

Line Circle Rose

Opens: Petals:

Max distance from pole:

Cycle:

θ	r
	5
	-2

6. $r = -8\sin(3\theta)$

Type:

Line Circle Rose

Opens: Petals:

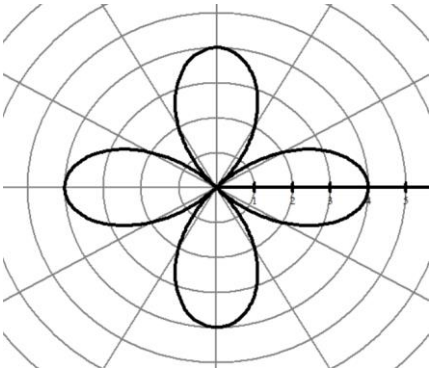
Max distance from pole:

Cycle:

θ	r
$\frac{\pi}{4}$	
$\frac{\pi}{2}$	

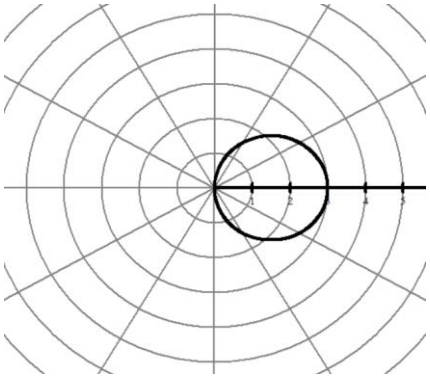
Write the equation of the polar function.

7.



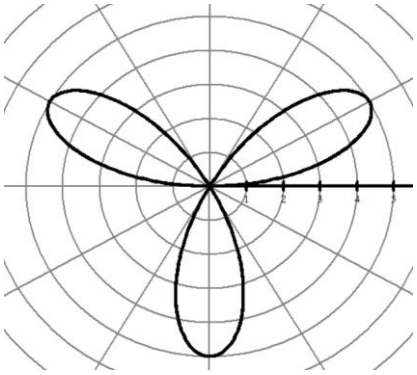
Equation:

8.



Equation:

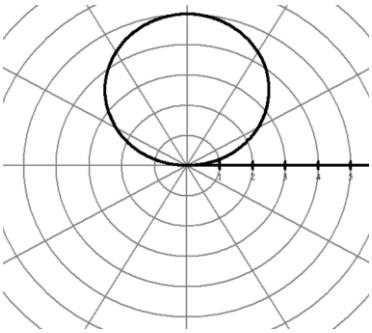
9.



Equation:

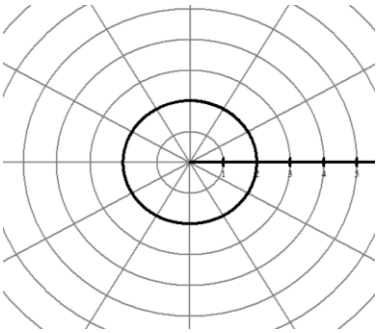
Write the equation of the polar function.

10.



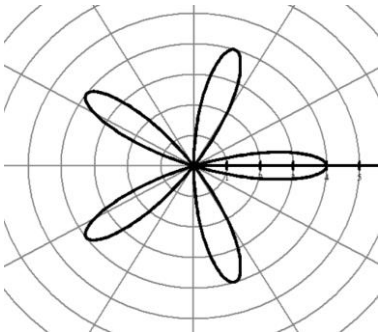
Equation:

11.



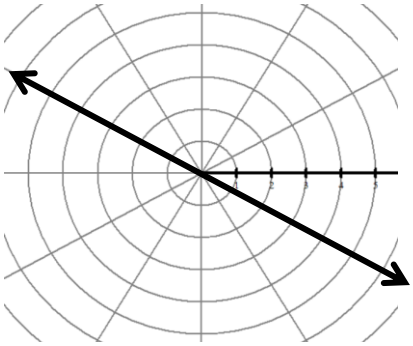
Equation:

12.



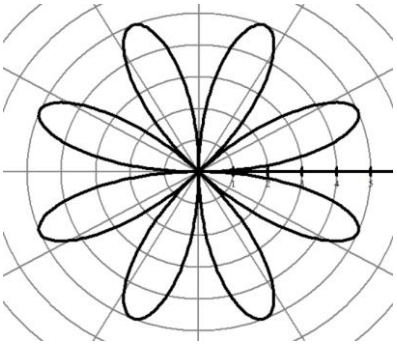
Equation:

13.



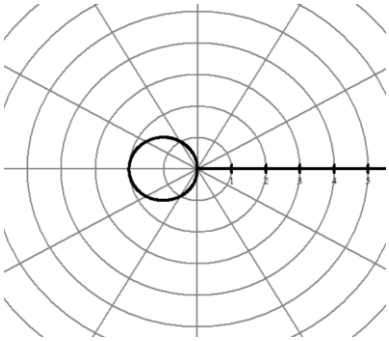
Equation:

14.



Equation:

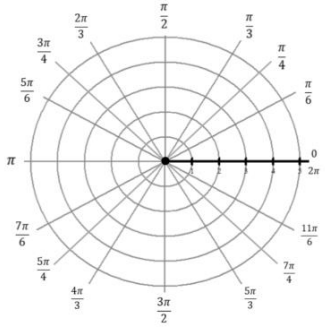
15.



Equation:

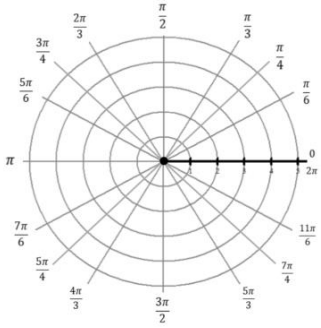
Sketch a graph. Find the endpoints of the restricted domain and highlight this interval on the function. (Use graphing calculator to verify your answers)

16. $r = 3$



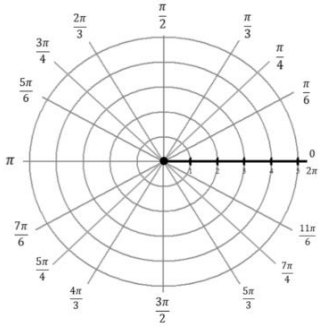
Endpoints of $\frac{\pi}{6} \leq \theta \leq \pi$

17. $r = 4\cos(3\theta)$



Endpoints of $\frac{\pi}{6} \leq \theta \leq \frac{\pi}{3}$

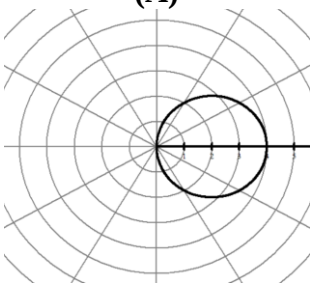
18. $r = 5\sin(\theta)$



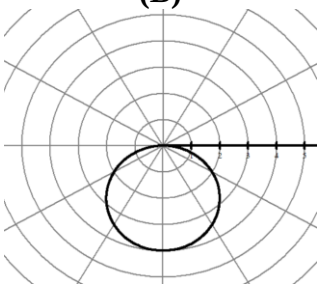
Endpoints of $\frac{\pi}{2} \leq \theta \leq \pi$

19. Which of the following is the graph of the polar function $r = f(\theta)$, where $f(\theta) = -4\cos\theta$, in the polar coordinate system for $0 \le \theta \le 2\pi$?

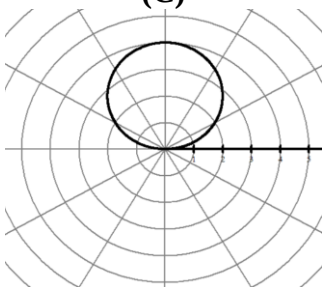
(A)



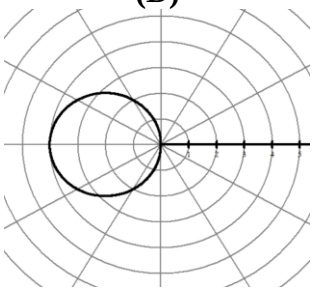
(B)



(C)



(D)



20. The graph of polar function $r = f(\theta)$ and $r = g(\theta)$, where $f(\theta) = 4\cos\theta$ and $g(\theta) = -4\sin\theta$, in the polar coordinate system for $0 \le \theta \le 2\pi$. Which of the following is a possible polar coordinate for $f(\theta) = g(\theta)$?

(A) $\left(2\sqrt{2},\ \frac{\pi}{4}\right)$

(B) $\left(2\sqrt{2},\ \frac{3\pi}{4}\right)$

(C) $\left(2\sqrt{2},\ \frac{5\pi}{4}\right)$

(D) $\left(2\sqrt{2},\ \frac{7\pi}{4}\right)$

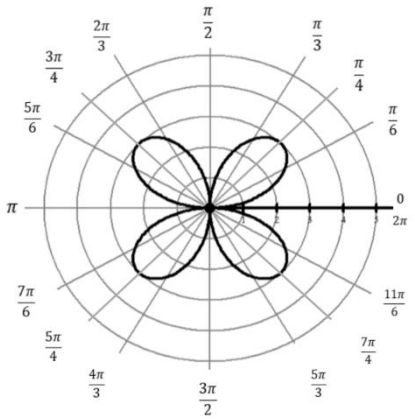
21. The graph of the polar function $r = f(\theta)$, is given the polar coordinate system. Which of the following defines $f(\theta)$ for $0 \le \theta \le 2\pi$?

(A) $f(\theta) = 3\sin(2\theta)$

(B) $f(\theta) = 3\sin(4\theta)$

(C) $f(\theta) = 3\cos(2\theta)$

(D) $f(\theta) = 3\cos(4\theta)$



3.14A Polar Function Graphs

AP Precalculus

Name: _____

CA #1

Use the polar function to answer the following. Fill in the table.

1. $r = 8\cos(4\theta)$

Type:

Line Circle Rose

Opens: Petals:

Cycle:

θ	r
$\frac{\pi}{6}$	
π	

2. $r = -7\sin(\theta)$

Type:

Line Circle Rose

Opens: Petals:

Cycle:

θ	r
$\frac{\pi}{3}$	
$\frac{\pi}{2}$	

3. $r = 7$

Type:

Line Circle Rose

Center: Petals:

Cycle:

θ	r
$\frac{\pi}{4}$	
$\frac{3\pi}{2}$	

Use the polar function to answer the following. Find the endpoints of the given domain.

4. $r = 6\sin(5\theta)$

Type:

Line Circle Rose

Opens: Petals:

Cycle:

Endpoints of $\frac{\pi}{6} \le \theta \le \frac{\pi}{3}$

5. $r = 9\cos(6\theta)$

Type:

Line Circle Rose

Opens: Petals:

Cycle:

Endpoints of $\frac{\pi}{3} \le \theta \le \frac{\pi}{2}$

6. $r = 8\cos(\theta)$

Type:

Line Circle Rose

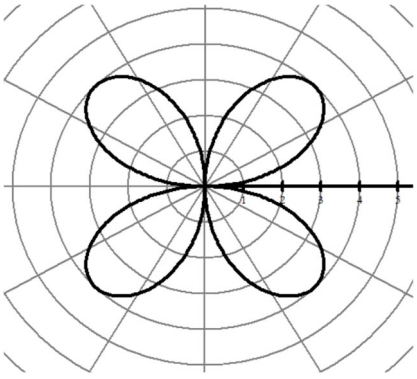
Opens: Petals:

Cycle:

Endpoints of $\frac{\pi}{4} \le \theta \le \frac{3\pi}{2}$

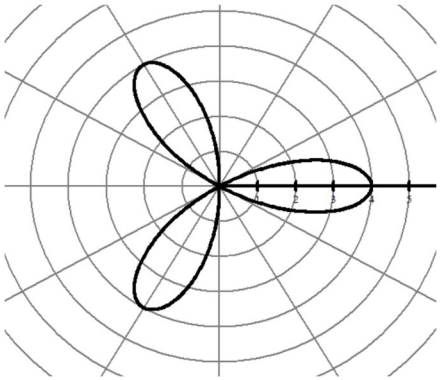
Write the equation of the polar function.

7.



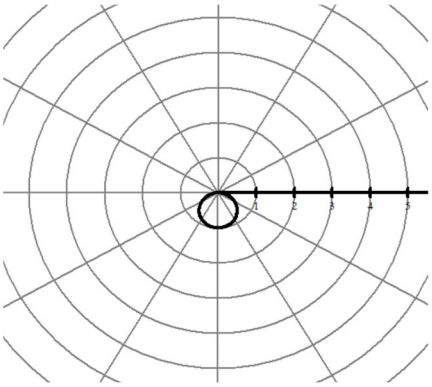
Equation:

8.



Equation:

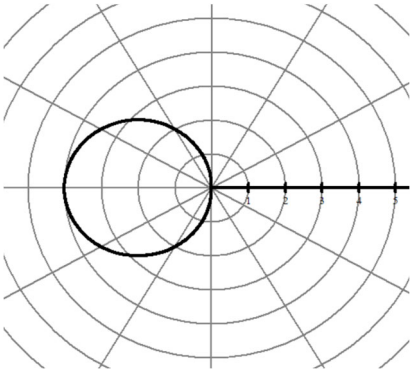
9.



Equation:

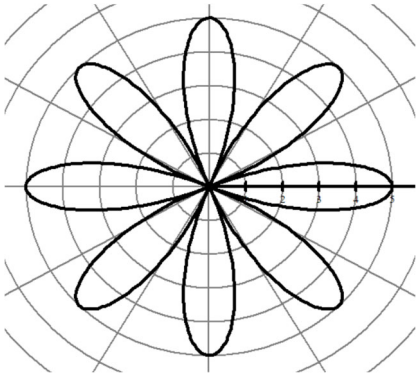
Write the equation of the polar function.

10.



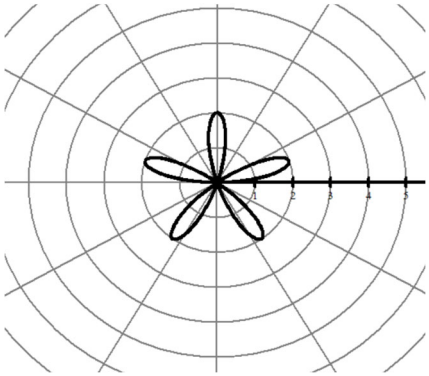
Equation:

11.



Equation:

12.



Equation:

Answers to 3.14A CA #1

1.
Rose with 8 petals
Cycle: $[0,2\pi]$

$\boldsymbol{\theta}$	$\boldsymbol{r}$
$\frac{\pi}{6}$	-4
π	8

2.
Circle opens down
Cycle: $[0,\pi]$

$\boldsymbol{\theta}$	$\boldsymbol{r}$
$\frac{\pi}{3}$	$-\frac{7\sqrt{3}}{2}$
$\frac{\pi}{2}$	-7

3.
Circle center is pole
Cycle: $[0,2\pi]$

$\boldsymbol{\theta}$	$\boldsymbol{r}$
$\frac{\pi}{4}$	7
$\frac{3\pi}{2}$	7

4.
Rose with 5 petals
Cycle: $[0,\pi]$

Endpoints: $\left(3,\frac{\pi}{6}\right), \left(-3\sqrt{3},\frac{\pi}{3}\right)$

5.
Rose with 12 petals
Cycle: $[0,2\pi]$

Endpoints: $\left(9,\frac{\pi}{3}\right), \left(-9,\frac{\pi}{2}\right)$

6.
Circle opens right
Cycle: $[0,\pi]$

Endpoints: $\left(4\sqrt{2},\frac{\pi}{4}\right), \left(0,\frac{3\pi}{2}\right)$

7.
Equation: $r = 4\sin(2\theta)$

8.
Equation: $r = 4\cos(3\theta)$

9.
Equation: $r = -\sin(\theta)$

10.
Equation: $r = -4\cos(\theta)$

11.
Equation: $r = 5\cos(4\theta)$

12.
Equation: $r = 2\sin(5\theta)$

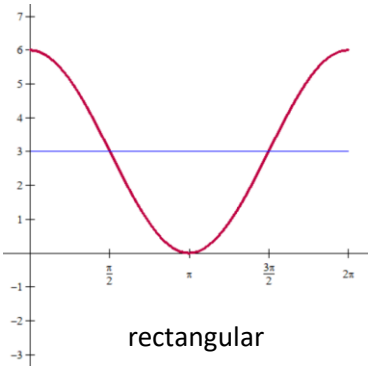
3.14B Polar Function Graphs

3.14B Notes

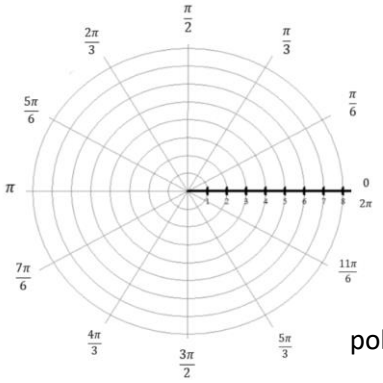
Limaçon

$$r = a \pm b\cos(\theta)$$

EXAMPLE: $r = 3 + 3\cos(\theta)$ $\quad a = b$ **CARDIOID**

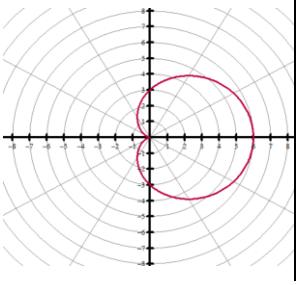


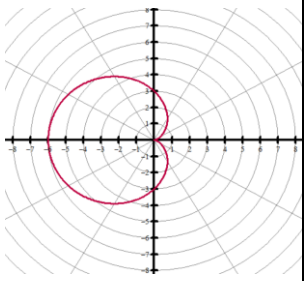
rectangular

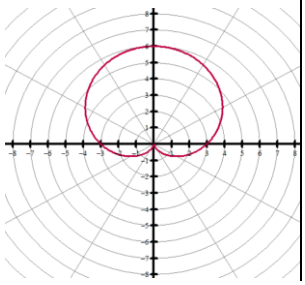


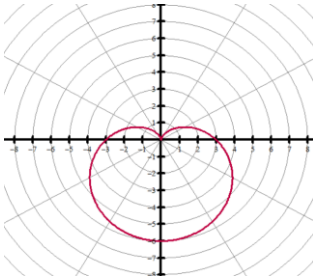
polar

$r = a + b\cos(\theta)$ | $r = a - b\cos(\theta)$ | $r = a + b\sin(\theta)$ | $r = a - b\sin(\theta)$

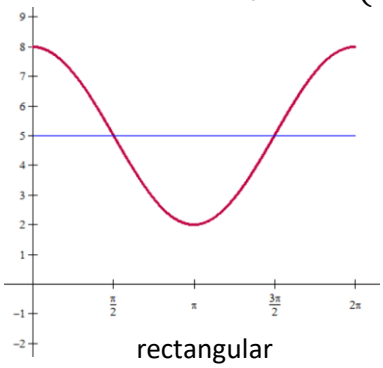




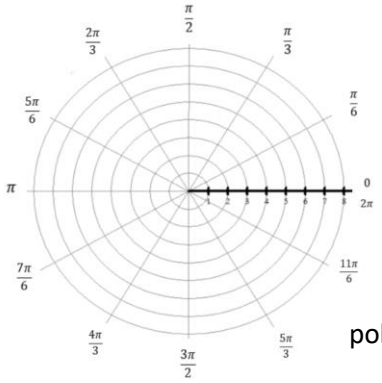




EXAMPLE: $r = 5 + 3\cos(\theta)$ $\quad a > b$ **DIMPLED CARDIOID**

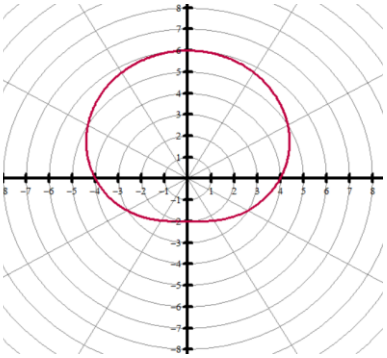


rectangular



polar

Use the graph of the polar function $r = f(\theta)$ for $0 \le \theta \le 2\pi$ to answer the following.



1. Write the equation of the polar function.

A	B	C	D
$r = 4 + 2\cos\theta$	$r = 4 - 2\cos\theta$	$r = 4 + 2\sin\theta$	$r = 4 - 2\sin\theta$

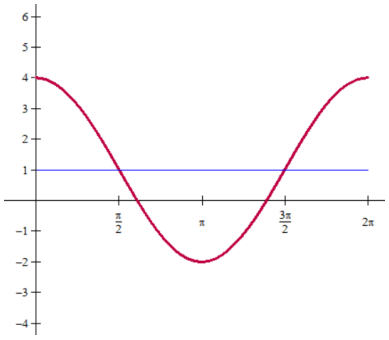
2. If the domain of f is restricted to $\frac{\pi}{2} \le \theta \le \pi$,

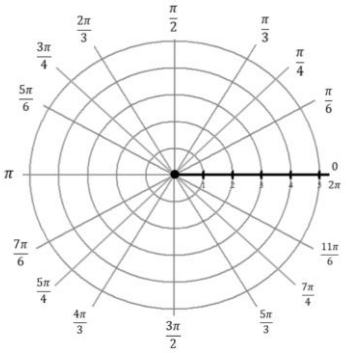
   a. Is the function positive or negative?

   b. Is the function increasing or decreasing?

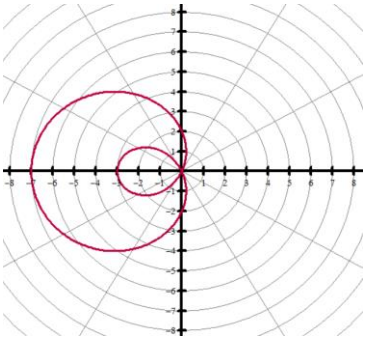
Write your questions and thoughts here!

EXAMPLE: $r = 1 + 3\cos(\theta)$ $\quad a < b$ **Inner Loop Limaçon**





Use the graph of the polar function $r = f(\theta)$ for $0 \le \theta \le 2\pi$ to answer the following.



1. Which is the equation of the polar function?

A	B	C	D
$r = 2 + 5\cos\theta$	$r = 2 - 5\cos\theta$	$r = 2 + 5\sin\theta$	$r = 2 - 5\sin\theta$

2. If the domain of f is restricted to $\frac{\pi}{2} \le \theta \le \pi$,

   a. Is the function positive or negative?

   b. Is the function increasing or decreasing?

Limaçon

$$r = a \pm b\cos(\theta)$$

$a > 0$ **and** $b > 0$

$a = b$ **cardioid**

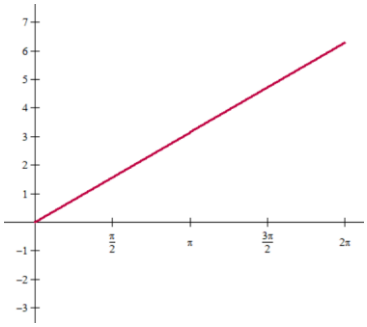
$a > b$, $1 < \frac{a}{b} < 2$ **dimpled cardioid**
(one loop limaçon)

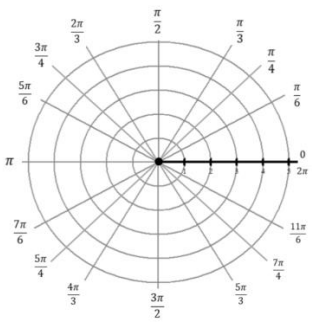
$a < b$ **inner loop limaçon**

$r = 5 + 7\sin(\theta)$	$r = 8 - 6\cos(\theta)$	$r = 6 + 6\sin(\theta)$

Archimedes' Spiral

EAMPLE: $r = \theta$ where $\theta \ge 0$





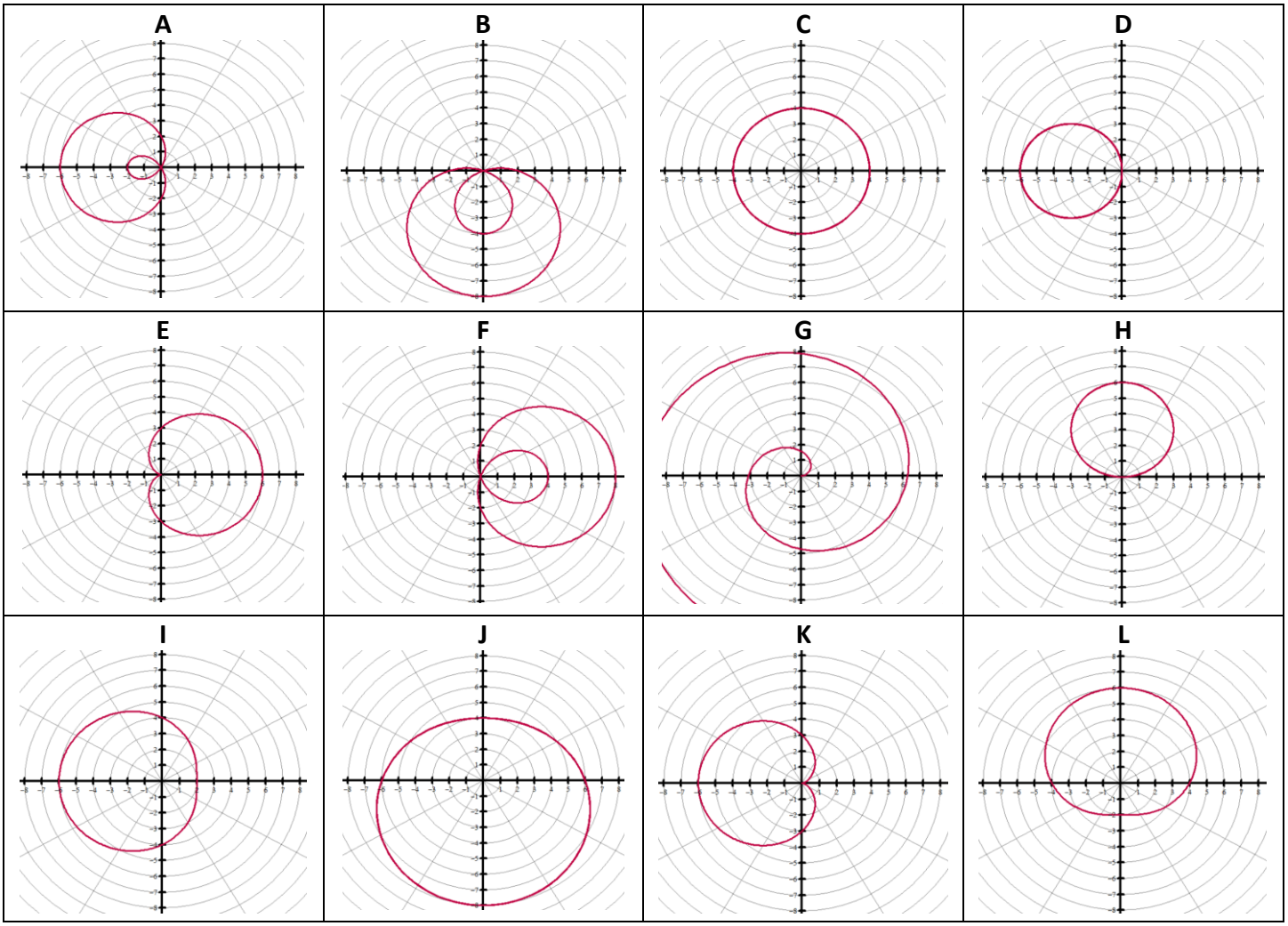
3.14B Polar Function Graphs

AP Precalculus

3.14B Practice

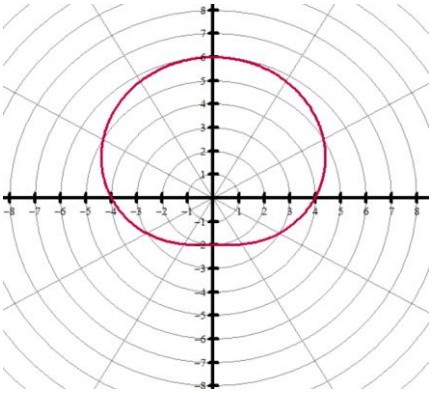
Match the equation to its graph below.		
1. $r = 2 + 6\cos(\theta)$ matches graph: _____	2. $r = 2 - 6\sin(\theta)$ matches graph: _____	3. $r = 3 + 3\cos(\theta)$ matches graph: _____
4. $r = 3 - 3\cos(\theta)$ matches graph: _____	5. $r = 2 - 4\cos(\theta)$ matches graph: _____	6. $r = 4 + 2\sin(\theta)$ matches graph: _____
7. $r = 4 - 2\cos(\theta)$ matches graph: _____	8. $r = \theta$ matches graph: _____	9. $r = -6\cos(\theta)$ matches graph: _____
10. $r = 6 - 2\sin(\theta)$ matches graph: _____	11. $r = 4$ matches graph: _____	12. $r = 6\sin(\theta)$ matches graph: _____

GRAPHS:



Circle the correct equation for the following polar graphs.

13.

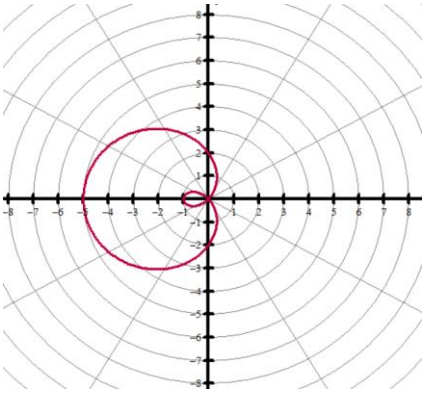


$r = 4 + 2\cos(\theta)$ $r = 4 + 2\sin(\theta)$

$r = 2 - 4\cos(\theta)$ $r = 2 - 4\sin(\theta)$

$r = 4 - 2\cos(\theta)$ $r = 4 - 2\sin(\theta)$

14.

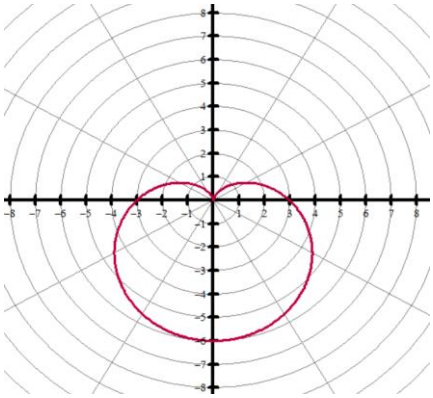


$r = 3 + 2\cos(\theta)$ $r = 3 + 2\sin(\theta)$

$r = 2 - 3\cos(\theta)$ $r = 2 - 3\sin(\theta)$

$r = 3 - 2\cos(\theta)$ $r = 3 - 2\sin(\theta)$

15.



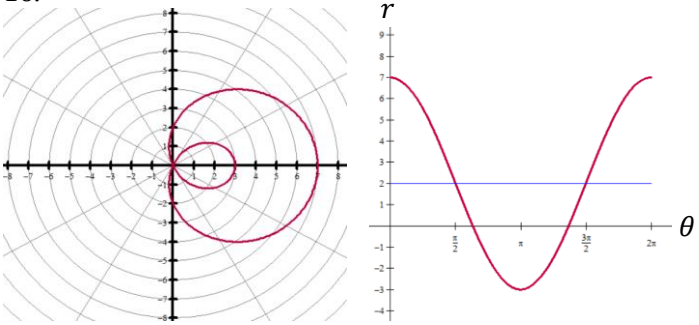
$r = 3 + 3\cos(\theta)$ $r = 3 + 3\sin(\theta)$

$r = 3 - 3\cos(\theta)$ $r = 3 - 3\sin(\theta)$

$r = 6 - 6\cos(\theta)$ $r = 6 - 6\sin(\theta)$

Use the graphs of the polar function $r = f(\theta)$ and $y = f(\theta)$ for $0 \le \theta \le 2\pi$ to answer the following.

16.

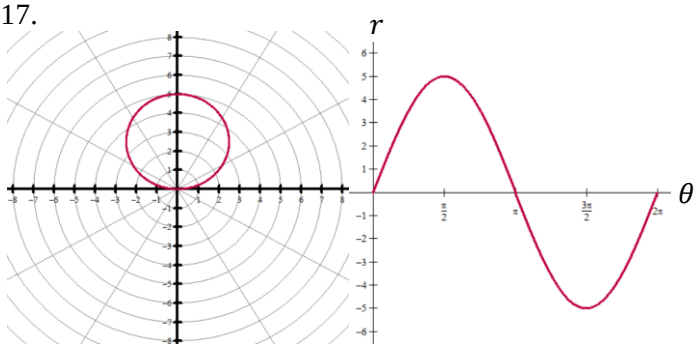


If the domain of f is restricted to $\frac{3\pi}{2} \le \theta \le 2\pi$,

a. Is the function positive or negative?

b. Is the function increasing or decreasing?

17.

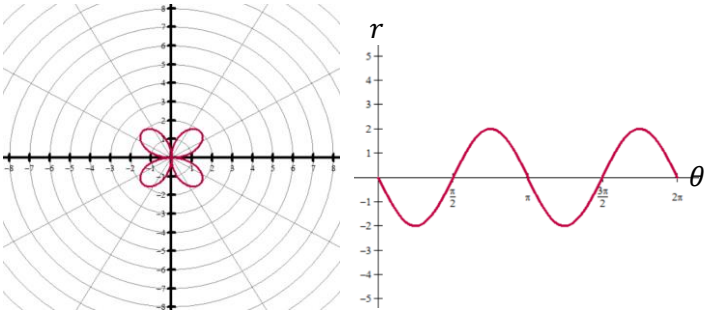


If the domain of f is restricted to $\pi \le \theta \le \frac{3\pi}{2}$,

a. Is the function positive or negative?

b. Is the function increasing or decreasing?

18.

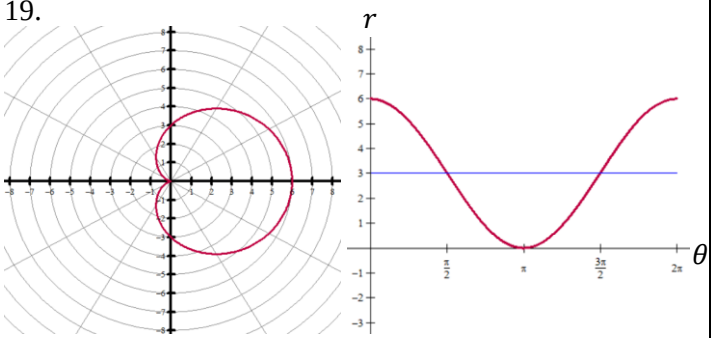


If the domain of f is restricted to $\frac{\pi}{4} \le \theta \le \frac{\pi}{2}$,

a. Is the function positive or negative?

b. Is the function increasing or decreasing?

19.



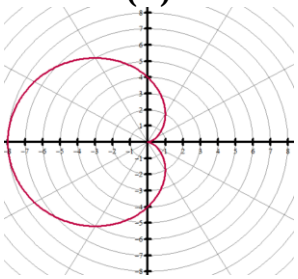
If the domain of f is restricted to $0 \le \theta \le \frac{\pi}{2}$,

a. Is the function positive or negative?

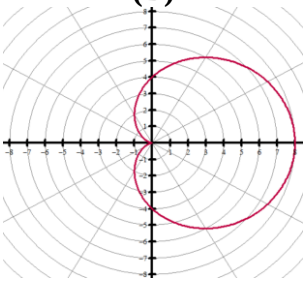
b. Is the function increasing or decreasing?

20. Which of the following is the graph of the polar function $r = f(\theta)$, where $f(\theta) = 4 - 4\cos\theta$, in the polar coordinate system for $0 \le \theta \le 2\pi$?

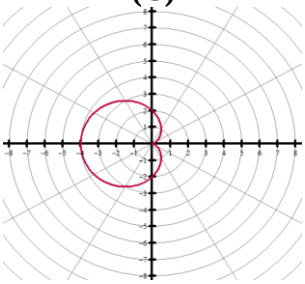
(A)



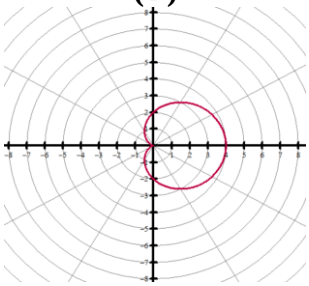
(B)



(C)



(D)



21. The polar function $r = f(\theta)$, where $f(\theta) = \frac{\pi}{3}\theta$, is defined for $\theta > 0$. Which of the following describes the graph of $r = f(\theta)$ in the polar coordinate system?

(A) The graph of $r = f(\theta)$ is line along the angle $\frac{\pi}{3}$

(B) The graph of $r = f(\theta)$ is a circle with radius of $\frac{\pi}{3}$

(C) The graph of $r = f(\theta)$ is a spiral with increasing radius

(D) The graph of $r = f(\theta)$ is a spiral with decreasing radius

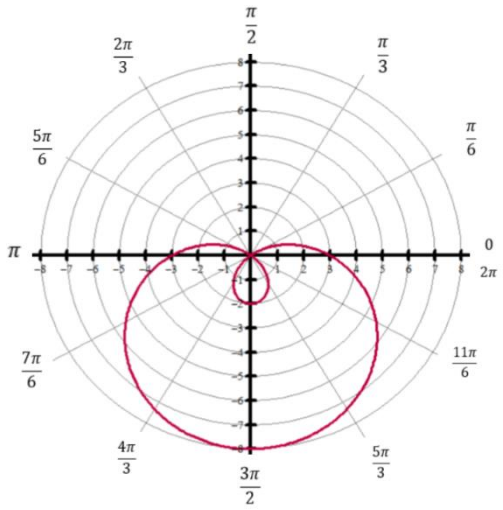
22. The graph of the polar function $r = f(\theta)$, is given the polar coordinate system. Which of the following defines $f(\theta)$ for $0 \le \theta \le 2\pi$?

(A) $f(\theta) = 3 - 5\sin(\theta)$

(B) $f(\theta) = 4 - 4\sin(\theta)$

(C) $f(\theta) = 2 - 2\sin(\theta)$

(D) $f(\theta) = 2 - 6\sin(\theta)$



3.14B Polar Function Graphs

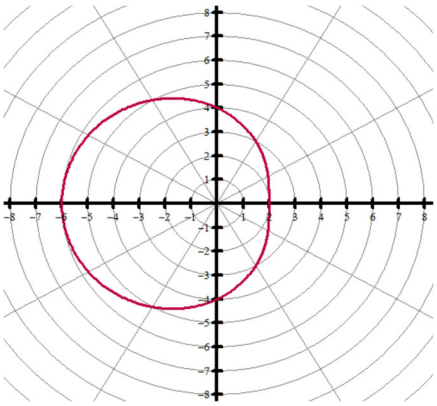
AP Precalculus

Name: _____

CA #1

Circle the correct equation for the following polar graphs.

1.



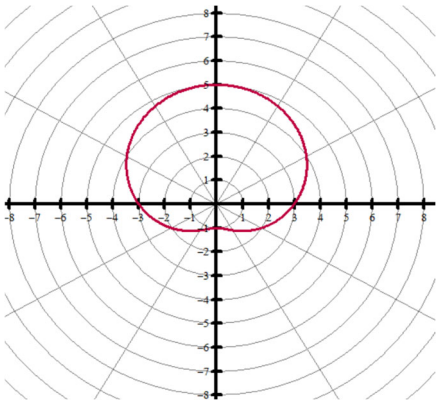
$r = 4 + 2\cos(\theta)$ $r = 4 + 2\sin(\theta)$

$r = 2 - 4\cos(\theta)$ $r = 2 - 4\sin(\theta)$

$r = 4 - 2\cos(\theta)$ $r = 4 - 2\sin(\theta)$

$r = 2 + 4\cos(\theta)$ $r = 2 + 4\sin(\theta)$

2.



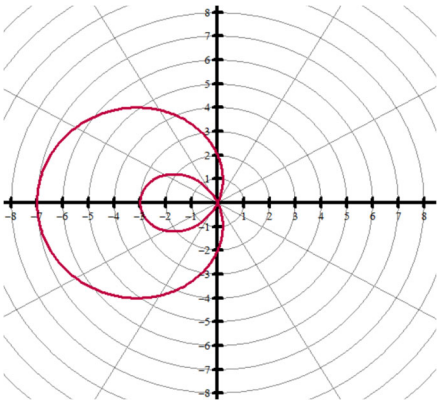
$r = 3 + 2\cos(\theta)$ $r = 3 + 2\sin(\theta)$

$r = 2 - 3\cos(\theta)$ $r = 2 - 3\sin(\theta)$

$r = 3 - 2\cos(\theta)$ $r = 3 - 2\sin(\theta)$

$r = 2 + 3\cos(\theta)$ $r = 2 + 3\sin(\theta)$

3.



$r = 3 + 4\cos(\theta)$ $r = 4 + 3\sin(\theta)$

$r = 3 - 4\cos(\theta)$ $r = 4 - 3\sin(\theta)$

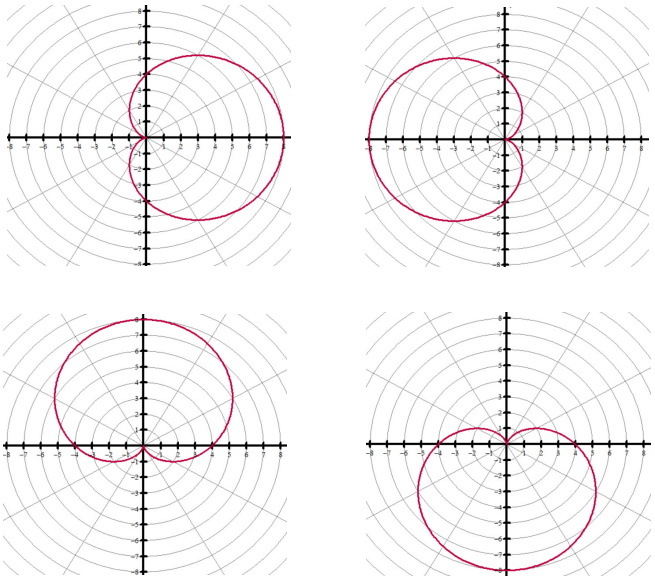
$r = 2 - 5\cos(\theta)$ $r = 2 - 5\sin(\theta)$

$r = 5 - 2\cos(\theta)$ $r = 5 - 2\sin(\theta)$

Circle the correct graph for the polar equation.

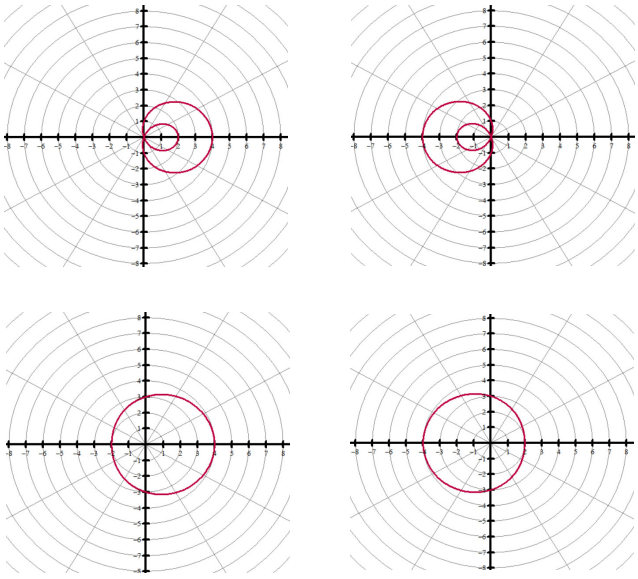
4.

$$r = 4 - 4\sin(\theta)$$



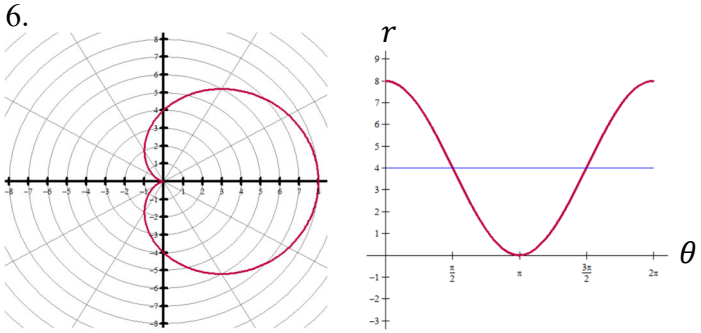
5.

$$r = 1 + 3\cos(\theta)$$



Use the graphs of the polar function $r = f(\theta)$ and $y = f(\theta)$ for $0 \le \theta \le 2\pi$ to answer the following.

6.

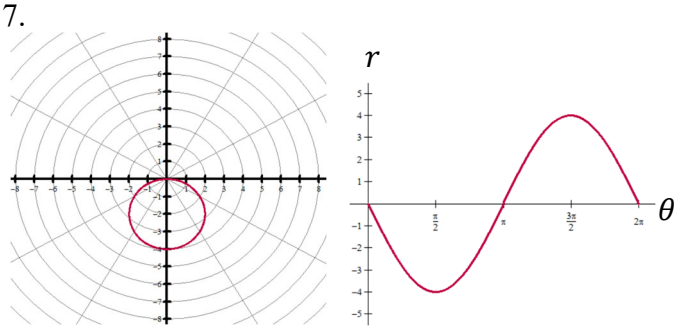


If the domain of f is restricted to $\frac{\pi}{2} \le \theta \le \pi$,

a. Is the function positive or negative?

b. Is the function increasing or decreasing?

7.

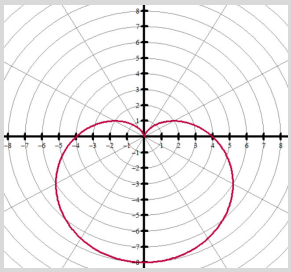
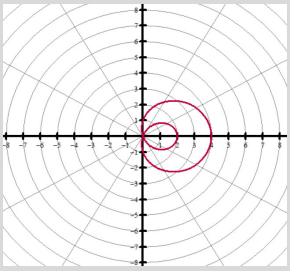


If the domain of f is restricted to $\frac{\pi}{2} \le \theta \le \pi$,

a. Is the function positive or negative?

b. Is the function increasing or decreasing?

Answers to 3.14B CA #1

1. $r = 4 - 2\cos(\theta)$	2. $r = 3 + 2\sin(\theta)$	3. $r = 2 - 5\cos(\theta)$
4. 	5. 	6. a. positive b. decreasing
7. a. negative b. increasing		

AP Precalc

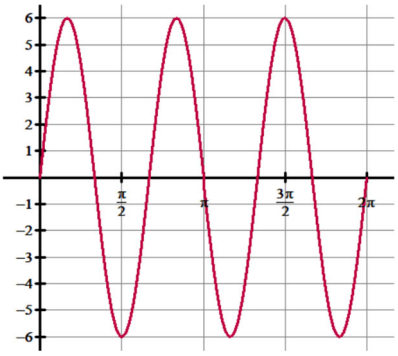
Write your questions and thoughts here!

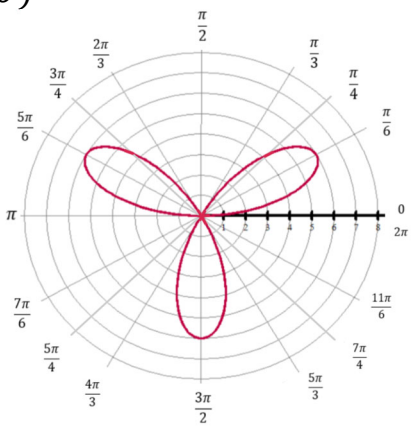
3.15 Rates of Change in Polar Functions

3.15 Notes

Rate of Change of r

EXAMPLE #1: $r = 6\sin(3\theta)$





***r* on the intervals:**

$\left(0, \frac{\pi}{6}\right)$

$\left(\frac{\pi}{6}, \frac{\pi}{3}\right)$

$\left(\frac{\pi}{3}, \frac{\pi}{2}\right)$

$\left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$

Distance from the Pole

	r is positive	r is negative
$r = f(\theta)$ is increasing		
$r = f(\theta)$ is decreasing		

EXAMPLE #2: $r = f(\theta)$

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{5\pi}{6}$	π
r	0	2.5	4.33	5	4.33	2.5	0

a. Describe the behavior of the function on the interval $0 \le \theta \le \frac{\pi}{2}$.

b. Describe the behavior of the function on the interval $\frac{\pi}{2} \le \theta \le \pi$.

c. What is the average rate of change on the interval $\frac{\pi}{6} \le \theta \le \frac{\pi}{3}$?

d. Estimate the value of $f\left(\frac{\pi}{4}\right)$.

Write your questions and thoughts here!

EXAMPLE #3: $r = f(\theta) = 3 + 5\sin(\theta)$ on the interval $\pi \leq \theta \leq 2\pi$

$\boldsymbol{\theta}$	π	$\frac{7\pi}{6}$	$\frac{5\pi}{4}$	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	$\frac{7\pi}{4}$	$\frac{11\pi}{6}$	2π
$\boldsymbol{r}$									

a. Determine the interval(s) where f is increasing.

b. Determine the interval(s) where f is decreasing.

c. Is there at least one relative extrema on the interval $\pi \leq \theta \leq 2\pi$? Explain.

d. The distance between $f(\theta)$ and the pole is increasing or decreasing on the interval $\frac{5\pi}{4} \leq \theta \leq \frac{3\pi}{2}$. Justify your answer.

e. The distance between $f(\theta)$ and the pole is increasing or decreasing on the interval $\pi \leq \theta \leq \frac{7\pi}{6}$. Justify your answer.

f. Find the average rate of change of f between $\theta = \frac{3\pi}{2}$ and $\theta = \frac{7\pi}{4}$.

g. On what interval(s) must $f(\theta) = 0$.

h. Compare the estimated value $f\left(\frac{5\pi}{3}\right)$ to the real value of $f\left(\frac{5\pi}{3}\right)$.

3.15 Rates of Change in Polar Functions

AP Precalculus

3.15 Practice

Use the table of selected values for the polar function $r = f(\theta)$ to answer the following.

1.

θ	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
r	11	9.53	6	2.46	1	2.46	6	9.53	11

a. Determine the interval(s) where f is increasing. Determine the interval(s) where f is decreasing.

b. The distance between $f(\theta)$ and the pole is increasing or decreasing on the interval $\pi \le \theta \le 2\pi$. Justify your answer.

c. Find the average rate of change of f between $\theta = \frac{5\pi}{4}$ and $\theta = \frac{7\pi}{4}$.

d. Estimate the value of $f\left(\frac{\pi}{3}\right)$ using an average rate of change.

e. Are there any extrema on the interval $\left[\frac{\pi}{4}, \frac{5\pi}{4}\right]$? Explain how you know.

2.

a. Is f increasing or decreasing on the interval $\frac{\pi}{4} \le \theta \le \frac{\pi}{2}$?

b. Is the distance between $f(\theta)$ and the pole is increasing or decreasing on the interval $0 \le \theta \le \frac{\pi}{4}$?

c. Is the rate of change of f faster on the interval $\left[0, \frac{\pi}{8}\right]$ or the interval $\left[\frac{\pi}{8}, \frac{\pi}{4}\right]$? Justify.

θ	r
0	0
$\frac{\pi}{8}$	-1.41
$\frac{\pi}{4}$	-2
$\frac{3\pi}{8}$	-1.41
$\frac{\pi}{2}$	0

Use the polar function $r = f(\theta)$ to fill in the table and answer the questions. Calculator Active.

3. $r = f(\theta) = 8\cos(\theta)$

$\boldsymbol{\theta}$	0	$\frac{\pi}{6}$	$\frac{\pi}{2}$	$\frac{5\pi}{6}$	π	$\frac{7\pi}{6}$	$\frac{3\pi}{2}$	$\frac{11\pi}{6}$	2π
$\boldsymbol{r}$									

a. Determine the interval(s) where f is increasing. Determine the interval(s) where f is decreasing.

b. How many extrema on the interval $\frac{5\pi}{6} \le \theta \le \frac{11\pi}{6}$?

c. Determine the intervals where the distance between $f(\theta)$ and the pole is increasing on the interval $0 \le \theta \le 2\pi$. Justify your answer.

d. Determine the intervals where the distance between $f(\theta)$ and the pole is decreasing on the interval $0 \le \theta \le 2\pi$. Justify your answer.

e. Find the average rate of change of f between $\theta = \frac{\pi}{2}$ and $\theta = \frac{5\pi}{6}$. Use to estimate $f\left(\frac{2\pi}{3}\right)$.

---

4. $r = f(\theta) = -3 + 5\sin(\theta)$

a. Is the distance between $f(\theta)$ and the pole is increasing or decreasing on the interval $\frac{\pi}{2} \le \theta \le \frac{3\pi}{4}$?

b. Find the average rate of change of f between $\theta = \frac{\pi}{4}$ and $\theta = \frac{\pi}{2}$.

c. Estimate the value of $f\left(\frac{5\pi}{6}\right)$ using an average rate of change.

$\boldsymbol{\theta}$	$\boldsymbol{r}$
0	
$\frac{\pi}{4}$	
$\frac{\pi}{2}$	
$\frac{3\pi}{4}$	
π	

5. Consider the graph of the polar function $r = f(\theta)$, where θ is increasing in the polar coordinate system on the interval $0 \le \theta \le 2\pi$. Given $f(\theta) < 0$ and decreasing on the interval $\pi \le \theta \le \frac{3\pi}{2}$ which of the following statements is true about the distance between the point with polar coordinates $(f(\theta), \theta)$ and the origin.

   (A) The distance is increasing for $0 \le \theta \le 2\pi$.

   (B) The distance is decreasing for $0 \le \theta \le 2\pi$.

   (C) The distance is increasing for $\pi \le \theta \le \frac{3\pi}{2}$.

   (D) The distance is decreasing for $\pi \le \theta \le \frac{3\pi}{2}$.

Use the table of selected values for the polar equation $r = f(\theta)$ below to answer questions 6 and 7.

CALCULATOR ACTIVE

θ	0	$\frac{\pi}{8}$	$\frac{\pi}{4}$
r	-1	$-\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$

6. The graph of the polar function $r = f(\theta)$, is given the polar coordinate system. Which of the following descriptions is true ?

   (A) As θ increasing from 0 to $\frac{\pi}{4}$, the polar function $r = f(\theta)$ is increasing, and the distance between the point $(f(\theta), \theta)$ on the curve and the origin is increasing.

   (B) As θ increasing from 0 to $\frac{\pi}{4}$, the polar function $r = f(\theta)$ is increasing, and the distance between the point $(f(\theta), \theta)$ on the curve and the origin is decreasing.

   (C) As θ increasing from 0 to $\frac{\pi}{4}$, the polar function $r = f(\theta)$ is decreasing, and the distance between the point $(f(\theta), \theta)$ on the curve and the origin is increasing.

   (D) As θ increasing from 0 to $\frac{\pi}{4}$, the polar function $r = f(\theta)$ is decreasing, and the distance between the point $(f(\theta), \theta)$ on the curve and the origin is decreasing.

7. If the value of $r = f\left(\frac{\pi}{8}\right)$ is estimated using the average rate of change of the function over the interval $0 \le \theta \le \frac{\pi}{4}$, which of the following is true?

   (A) The estimated value would be an overestimate of the actual vale by approximately 0.116.

   (B) The estimated value would be an underestimate of the actual vale by approximately -0.23.

   (C) The estimated value would be an overestimate of the actual vale by approximately 1.616.

   (D) The estimated value would be an underestimate of the actual vale by approximately -1.043.

3.15 Rates of Change in Polar Functions

AP Precalculus Name: _____ **CA #1**

Use the table of selected values for the polar function $r = f(\theta)$ to answer the following.

1.

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	3	-1.24	-3	-1.24	3	7.24	9	7.24	3

a. Determine the interval(s) where f is increasing.

b. Determine the interval(s) where f is decreasing.

c. Are there any extrema on the interval $0 \le \theta \le \frac{7\pi}{4}$? Explain how you know.

d. Determine the interval(s) where distance between $f(\theta)$ and the pole is increasing on $0 \le \theta \le 2\pi$? Justify your work.

e. Determine the interval(s) where distance between $f(\theta)$ and the pole is decreasing on $0 \le \theta \le 2\pi$? Justify your work.

f. Find the average rate of change of f between $\theta = \pi$ and $\theta = \frac{5\pi}{4}$.

g. Estimate the value of $f\left(\frac{\pi}{3}\right)$ using an average rate of change.

Use the polar function $r = f(\theta)$ to fill in the table and answer the questions. Calculator Active.

2. $r = f(\theta) = 8\sin(2\theta)$

θ	r
0	
$\frac{\pi}{6}$	
$\frac{\pi}{4}$	

a. Is f increasing or decreasing on the interval $0 \le \theta \le \frac{\pi}{4}$?

b. Is the distance between $f(\theta)$ and the pole is increasing or decreasing on the interval $0 \le \theta \le \frac{\pi}{4}$?

c. Find the average rate of change of f between $\theta = \frac{\pi}{6}$ and $\theta = \frac{\pi}{4}$.

d. Estimate the value of $f\left(\frac{\pi}{6}\right)$ using an average rate of change.

Answers to 3.15 CA #1

1.

a. $\left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$

b. $\left(0, \frac{\pi}{2}\right)$ and $\left(\frac{3\pi}{2}, 2\pi\right)$

c. at least 2 extrema: changes from decreasing to increasing to decreasing

d. $\left(\pi, \frac{3\pi}{2}\right)$ r is positive and increasing

$\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$ r is negative and decreasing

e. $\left(\frac{3\pi}{2}, 2\pi\right)$ r is positive and decreasing

$\left(\frac{\pi}{2}, \frac{3\pi}{4}\right)$ r is negative and increasing

f. $\frac{16.96}{\pi} \approx 5.398$ units per radian

g. $y + 3 = -2.24\left(x - \frac{\pi}{2}\right)$

$f\left(\frac{\pi}{3}\right) \approx -1.827$

2.

θ	r
0	0
$\frac{\pi}{6}$	6.928
$\frac{\pi}{4}$	8

a. increasing

b. increasing: r is positive and increasing

c. $\frac{-12.864}{-\pi} \approx 4.09$ units per radian

d. $y - 0 = 10.18(x - 0)$

$f\left(\frac{\pi}{6}\right) \approx 5.33$

Name: _____ Date:_____ Period: _____ **3B Review**

Unit 3-B Review – Trigonometric Functions

Reviews do NOT cover all material from the lessons but will hopefully remind you of key points. To be prepared, you must study all packets for lessons 3.8 – 3.15.

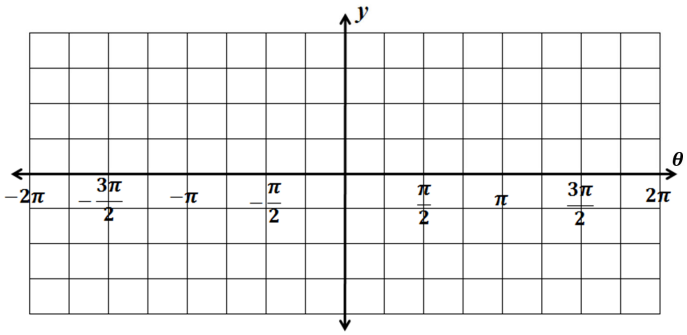
Write an equation that represents all asymptotes of the graph of f in the xy-plane.	**In the xy-plane, the angle θ is in standard position. What is the slope of the terminal ray of the angle?**
1. Let $f(\theta) = \tan\left(\frac{\theta}{6}\right)$.	2. $\theta = \frac{7\pi}{4}$

Evaluate.

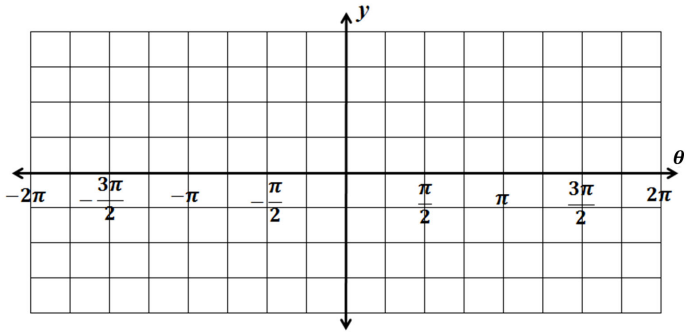
3. $\tan\frac{\pi}{6}$	4. $\tan\frac{3\pi}{2}$	5. $\tan\frac{4\pi}{3}$

Graph each trig function.

6. $g(\theta) = \tan\left(\theta + \frac{\pi}{4}\right) - 1$

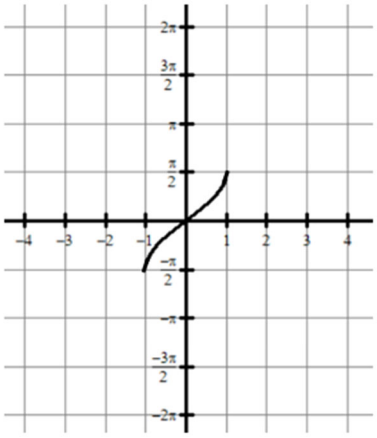


7. $g(\theta) = -2\tan(2\theta) + 3$

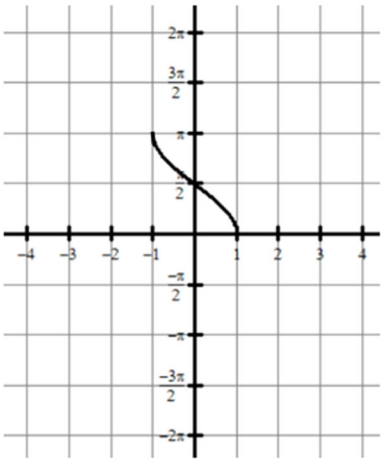


The parent function is shown below. Use the parent function to graph $g(x)$.

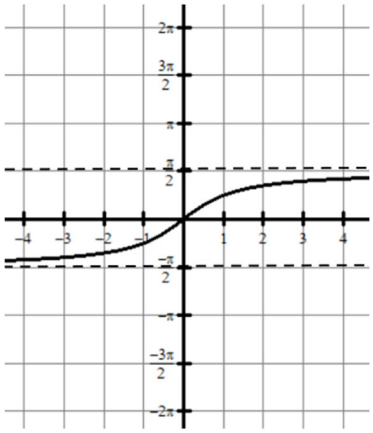
8. $g(x) = 3\sin^{-1}(-x) + \frac{\pi}{2}$



9. $g(x) = 2\cos^{-1}(x-3)$



10. $g(x) = -\tan^{-1}(x) - \pi$



Find the inverse of each function and list the domain and range of $f^{-1}(x)$.

11. $f(x) = 2\sin x - 5$ for $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$

Solve each equation for $0 \le \theta \le 2\pi$. Find the exact value(s) using the unit circle

12. $2\sin\theta + 5 = 4$

13. $\cos^2\theta + \cos\theta = 0$

Solve each equation. Find ALL approximate value(s) using a calculator.

14. $5 + 5\sin^2 x = 8$

15. $\tan^2 x - 3\tan x = 18$

Evaluate the following expressions. Use exact values.

16. $\sec\left(\frac{\pi}{3}\right)$

17. $\csc\left(-\frac{\pi}{2}\right)$

18. $\cot\left(\frac{5\pi}{6}\right)$

Evaluate the following expressions. Use approximate values from calculator.

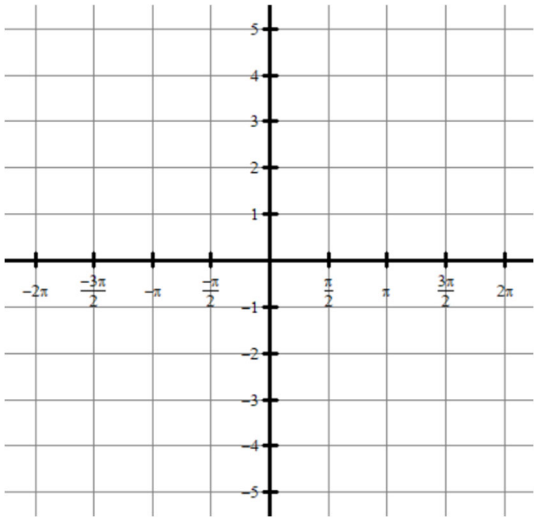
19. $\csc(1.84)$

20. $\sec\left(\frac{3\pi}{5}\right)$

21. $\cot\left(\frac{4\pi}{9}\right)$

Graph the following. State the range and all vertical asymptotes.

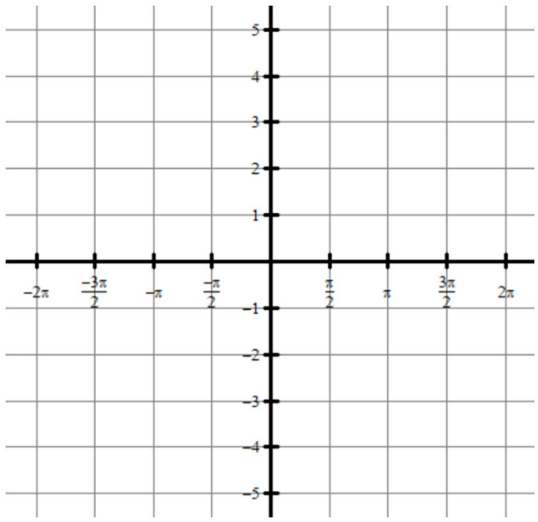
22. $f(x) = 2\csc\left(x + \frac{\pi}{2}\right) - 1$



Range:

Vertical Asymptotes:

23. $f(x) = 3\sec\left(\frac{1}{2}x\right)$



Range:

Vertical Asymptotes:

Use trig identities to write each expression in terms of a single trig identity.

24. $\cos^2 x \sec x$

25. $(1 - \sin^2 x)\csc^2 x$

26. $\frac{1}{1 + \cot^2 x}$

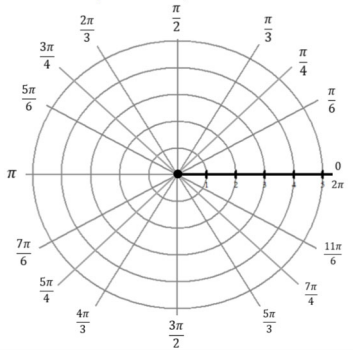
Find the exact value of the sum or difference.

27. $\sin\left(\frac{3\pi}{4} - \frac{5\pi}{6}\right)$

28. $\cos\left(\frac{\pi}{2} + \frac{2\pi}{3}\right)$

Polar Coordinates

29. Plot $\left(-3, \frac{2\pi}{3}\right)$



30. Convert from polar to rectangular.

$\left(4, \frac{5\pi}{6}\right)$

31. Convert from rectangular to polar where $0 \leq \theta \leq 2\pi$.

$(2, -4)$

32. Convert from rectangular complex to polar.

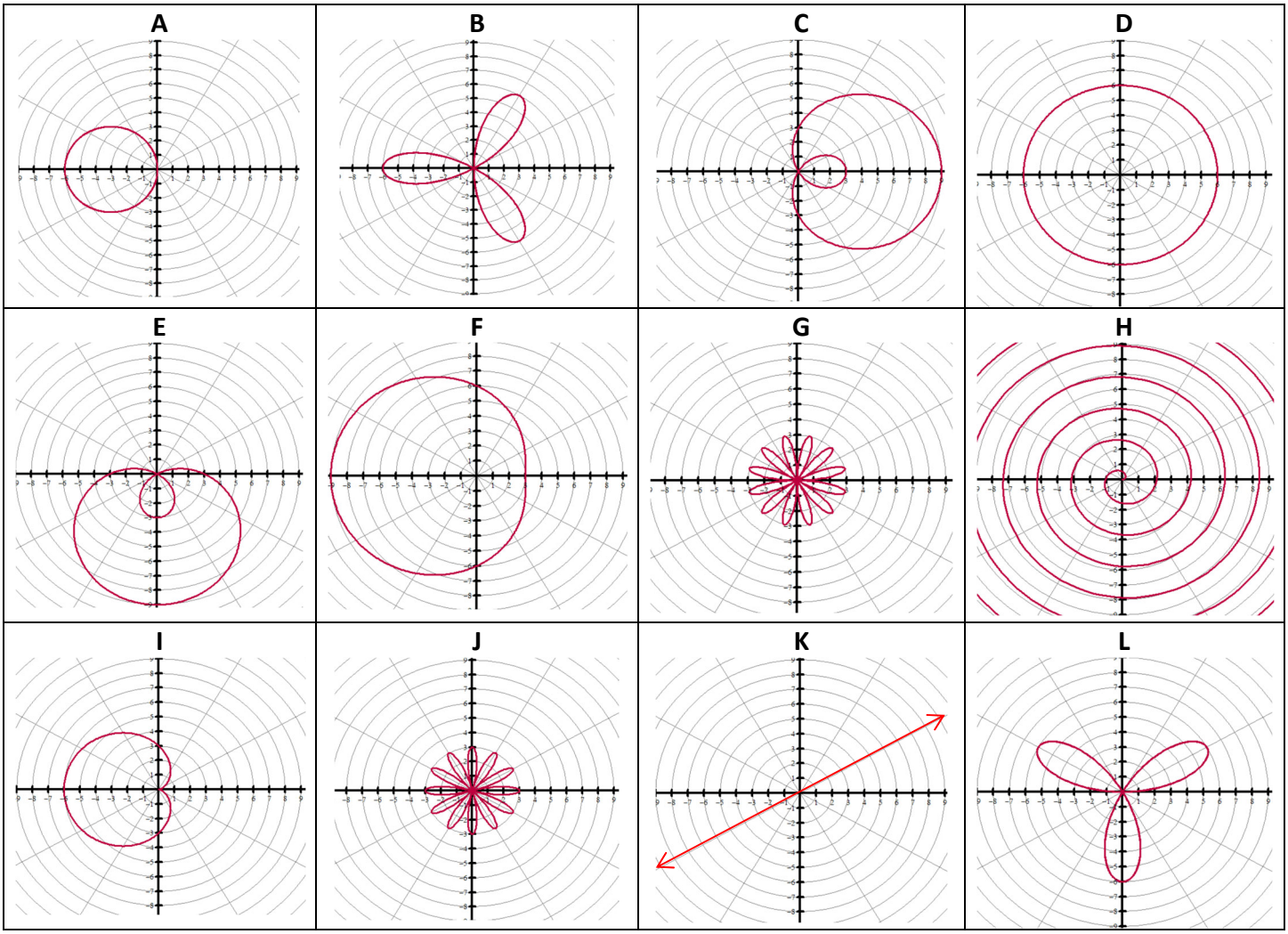
$-3 - 5i$

33. Convert from polar complex to rectangular.

$6\left[\cos\left(\frac{3\pi}{4}\right) + i\sin\left(\frac{3\pi}{4}\right)\right]$

Match the equation to its graph below.		
34. $r = 3 + 6\cos(\theta)$ matches graph: _____	35. $r = 3 - 6\sin(\theta)$ matches graph: _____	36. $r = 3 - 3\cos(\theta)$ matches graph: _____
37. $r = 6 - 3\cos(\theta)$ matches graph: _____	38. $r = -6\cos(\theta)$ matches graph: _____	39. $r = 6\sin(3\theta)$ matches graph: _____
40. $r = 3\cos(6\theta)$ matches graph: _____	41. $r = \frac{\theta}{3}$ matches graph: _____	42. $r = -6\cos(3\theta)$ matches graph: _____
43. $r = 3\sin(6\theta)$ matches graph: _____	44. $r = 6$ matches graph: _____	45. $\theta = \frac{\pi}{6}$ matches graph: _____

GRAPHS:



Use the table of selected values for the polar function $r = f(\theta)$ to answer the following.

46.

θ	r
0	-5
$\frac{\pi}{8}$	-4.619
$\frac{\pi}{4}$	-3.536
$\frac{3\pi}{8}$	-1.913
$\frac{\pi}{2}$	0

a. Is f increasing or decreasing on the interval $0 \le \theta \le \frac{\pi}{2}$?

b. Is the distance between $f(\theta)$ and the pole is increasing or decreasing on the interval $0 \le \theta \le \frac{\pi}{2}$?

c. Find the average rate of change of f between $\theta = \frac{\pi}{8}$ and $\theta = \frac{3\pi}{8}$.

d. Estimate the value of $f\left(\frac{\pi}{4}\right)$ using an average rate of change.

Multiple Choice

47. The function f is given by $f(x) = 4\sin(x) + 1$. For what values of x where $0 \le x \le 2\pi$ is $f(x) \le -1$?

(A) $\frac{7\pi}{6} \le x \le \frac{11\pi}{6}$

(B) $\frac{4\pi}{3} \le x \le \frac{5\pi}{3}$

(C) $0 \le x \le \frac{7\pi}{6}$ and $\frac{11\pi}{6} \le x \le 2\pi$

(D) $0 \le x \le \frac{4\pi}{3}$ and $\frac{5\pi}{3} \le x \le 2\pi$

Name: _____ Date:_____

Corrective Assignment

Unit 3-B – Trigonometric Functions

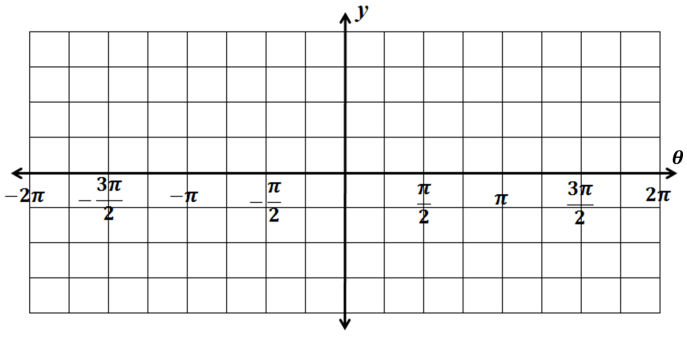
Write an equation that represents all asymptotes of the graph of f in the xy-plane.	**In the xy-plane, the angle θ is in standard position. What is the slope of the terminal ray of the angle?**
1. Let $f(\theta) = \tan(2\theta) + 1$.	2. $\theta = \frac{5\pi}{4}$

Evaluate.

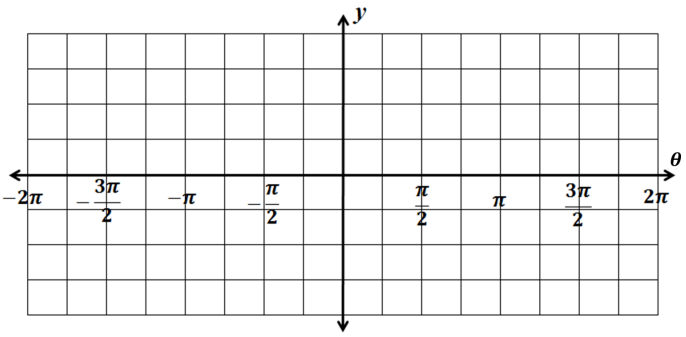
3. $\tan\frac{\pi}{3}$	4. $\tan\frac{\pi}{2}$	5. $\tan\frac{5\pi}{6}$

Graph each trig function.

6. $g(\theta) = 2\tan\left(\theta + \frac{\pi}{2}\right) - 1$

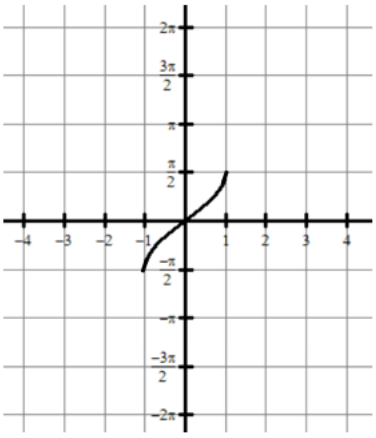


7. $g(\theta) = -\tan\left(\frac{\theta}{2}\right) + 2$

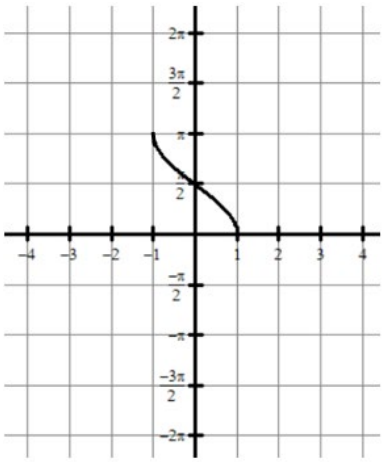


The parent function is shown below. Use the parent function to graph $g(x)$.

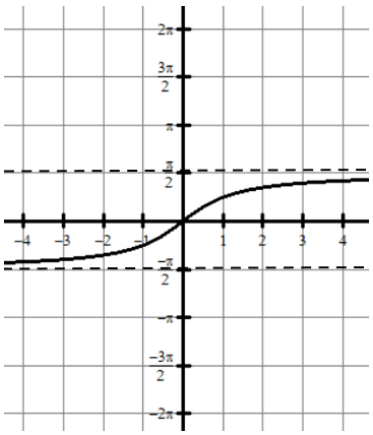
8. $g(x) = 2\sin^{-1}(-x) + \frac{\pi}{2}$



9. $g(x) = -\cos^{-1}(x+1)$



10. $g(x) = \tan^{-1}(x) + \pi$



Find the inverse of each function and list the domain and range of $f^{-1}(x)$.

11. $f(x) = 3\sin x - 5$ for $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$

Solve each equation for $0 \le \theta \le 2\pi$. Find the exact value(s) using the unit circle

12. $2\sin\theta + 6 = 4$

13. $\sin^2\theta = \sin\theta$

Solve each equation. Find ALL approximate value(s) using a calculator.

14. $1 + 5\sin x = 4$

15. $\tan^2 x - 3\tan x - 10 = 0$

Evaluate the following expressions. Use exact values.

16. $\sec\left(\frac{\pi}{6}\right)$

17. $\csc\left(\frac{3\pi}{2}\right)$

18. $\cot\left(\frac{4\pi}{3}\right)$

Evaluate the following expressions. Use approximate values from calculator.

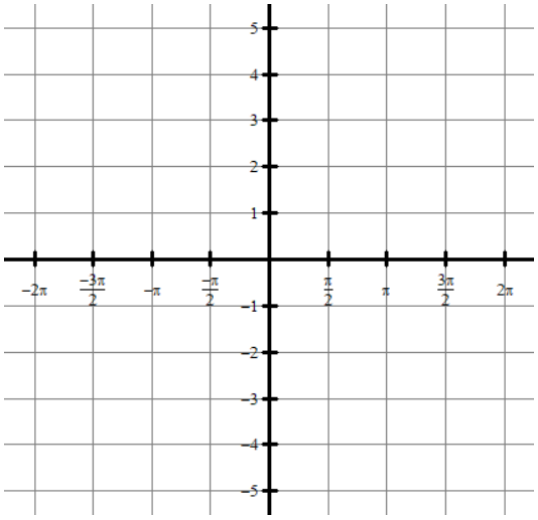
19. $\csc(1.24)$

20. $\sec\left(\frac{4\pi}{5}\right)$

21. $\cot\left(\frac{4\pi}{7}\right)$

Graph the following. State the range and all vertical asymptotes.

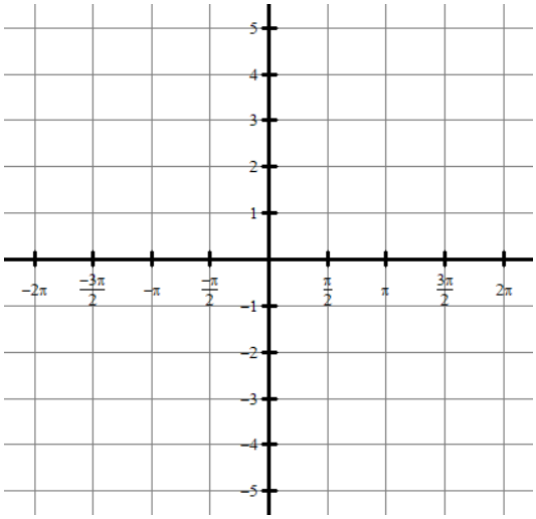
22. $f(x) = 3\csc(2x) - 1$



Range:

Vertical Asymptotes:

23. $f(x) = 2\sec\left(\frac{1}{2}(x+\pi)\right)$



Range:

Vertical Asymptotes:

Use trig identities to write each expression in terms of a single trig identity.

24. $\csc^2 x \sin x$

25. $(1 - \sin^2 x)\tan^2 x$

26. $\frac{\sin^2 x + \cos^2 x}{\sin x}$

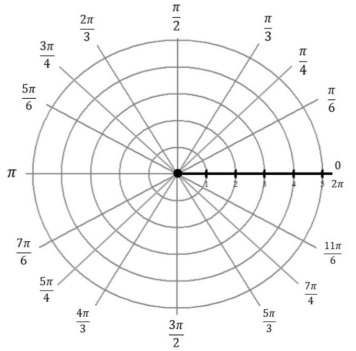
Find the exact value of the sum or difference.

27. $\sin\left(\frac{\pi}{4} - \frac{5\pi}{6}\right)$

28. $\cos\left(\frac{\pi}{2} + \frac{4\pi}{3}\right)$

Polar Coordinates

29. Plot $\left(3, -\frac{2\pi}{3}\right)$



30. Convert from polar to rectangular.

$\left(5, \frac{5\pi}{6}\right)$

31. Convert from rectangular to polar where $0 \le \theta \le 2\pi$.

$(-2, -4)$

32. Convert from rectangular complex to polar.

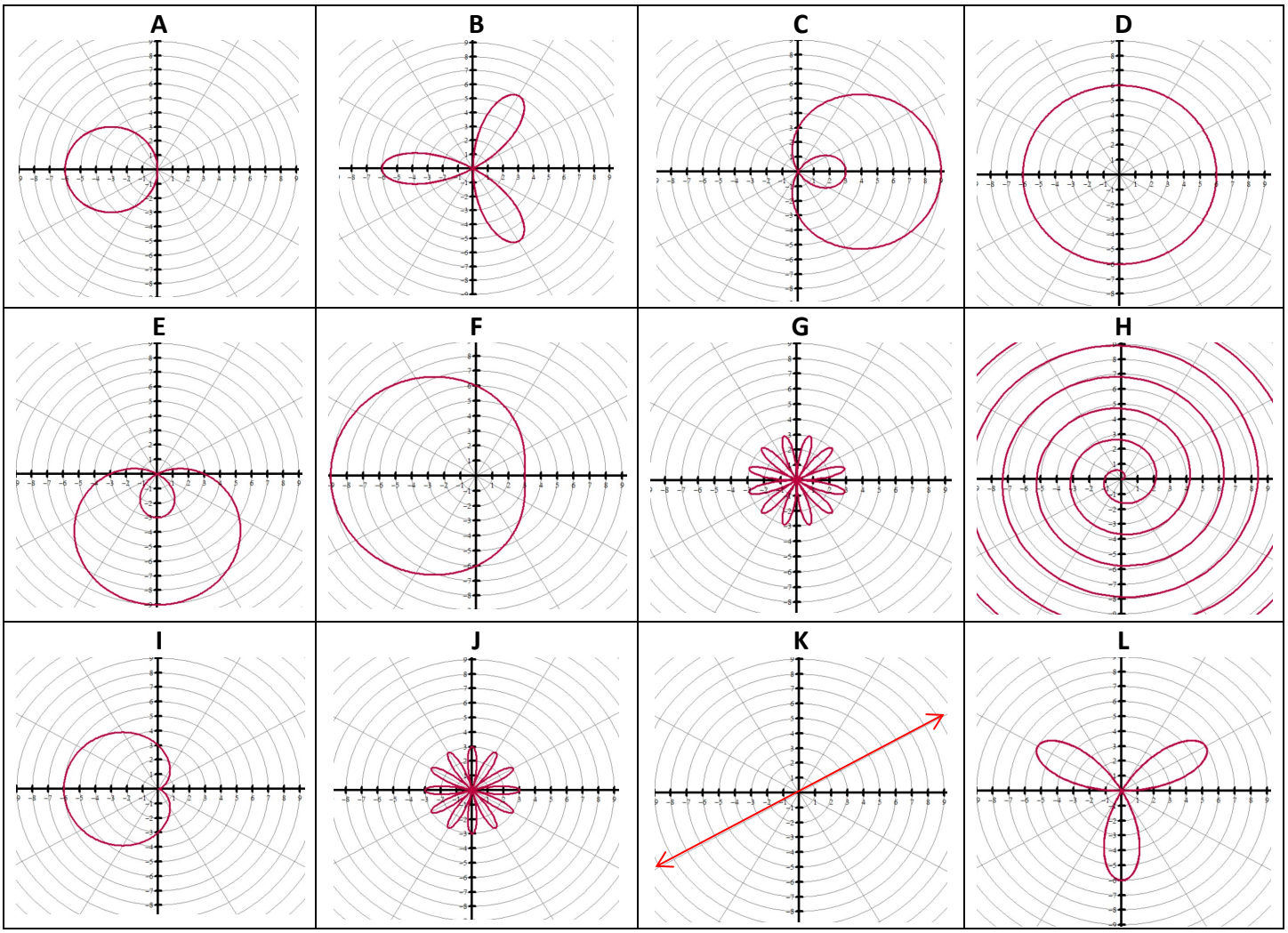
$-3 + 5i$

33. Convert from polar complex to rectangular.

$6\left[\cos\left(\frac{\pi}{4}\right) + i\sin\left(\frac{\pi}{4}\right)\right]$

Match the equation to its graph below.		
34. $r = 3 + 6\cos(\theta)$ matches graph: _____	35. $r = 3 - 6\sin(\theta)$ matches graph: _____	36. $r = 3 - 3\cos(\theta)$ matches graph: _____
37. $r = 6 - 3\cos(\theta)$ matches graph: _____	38. $r = -6\cos(\theta)$ matches graph: _____	39. $r = 6\sin(3\theta)$ matches graph: _____
40. $r = 3\cos(6\theta)$ matches graph: _____	41. $r = \frac{\theta}{3}$ matches graph: _____	42. $r = -6\cos(3\theta)$ matches graph: _____
43. $r = 3\sin(6\theta)$ matches graph: _____	44. $r = 6$ matches graph: _____	45. $\theta = \frac{\pi}{3}$ matches graph: _____

GRAPHS:



Use the table of selected values for the polar function $r = f(\theta)$ to answer the following.

46.

θ	r
0	3
$\frac{\pi}{8}$	4.913
$\frac{\pi}{4}$	6.535
$\frac{3\pi}{8}$	7.619
$\frac{\pi}{2}$	8

a. Is f increasing or decreasing on the interval $0 \leq \theta \leq \frac{\pi}{2}$?

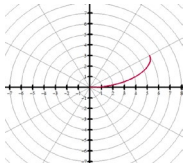
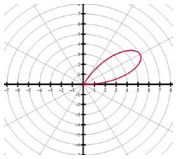
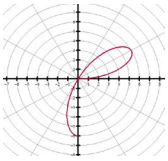
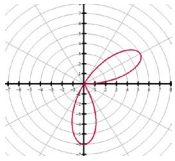
b. Is the distance between $f(\theta)$ and the pole is increasing or decreasing on the interval $0 \leq \theta \leq \frac{\pi}{2}$?

c. Find the average rate of change of f between $\theta = \frac{\pi}{8}$ and $\theta = \frac{3\pi}{8}$.

d. Estimate the value of $f\left(\frac{\pi}{4}\right)$ using an average rate of change.

Use the graph of the polar function over each interval to complete the table.

47.

Graph of $r = f(x)$				
Interval	$0 \leq x \leq \frac{\pi}{6}$	$\frac{\pi}{6} \leq x \leq \frac{\pi}{3}$	$\frac{\pi}{3} \leq x \leq \frac{\pi}{2}$	$\frac{\pi}{2} \leq x \leq \frac{2\pi}{3}$
r is Pos/Neg				
Distance from the origin Inc/Dec				

Answers to Unit 3B Corrective Assignment

1. $\frac{\pi}{4}+\frac{\pi}{2}n$ where n is an integer

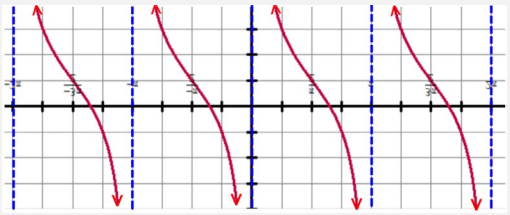
2. 1

3. $\sqrt{3}$

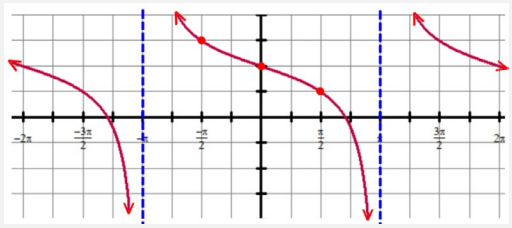
4. DNE

5. $-\frac{\sqrt{3}}{3}$

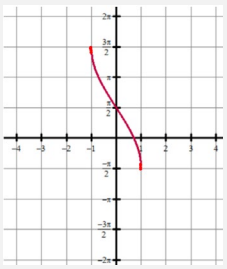
6.



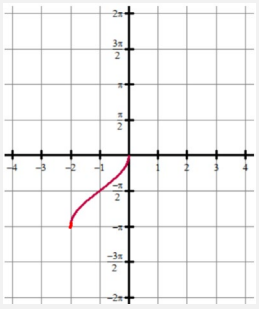
7.



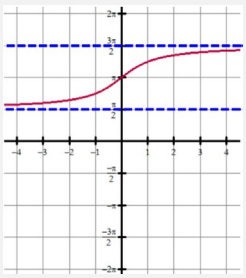
8.



9.



10.



11. $f^{-1}(x)=\sin^{-1}\left(\frac{x+5}{3}\right)$

Domain: $-8\le x\le -2$

Range: $-\frac{\pi}{2}\le y\le \frac{\pi}{2}$

12. $\theta=\frac{3\pi}{2}$

13. $\theta=\frac{\pi}{2},\pi,2\pi$

14. $x=0.643$
$x=2.498$

15. $x=2.034$ and 5.176
$x=1.373$ and 4.514

16. $\frac{2\sqrt{3}}{3}$

17. -1

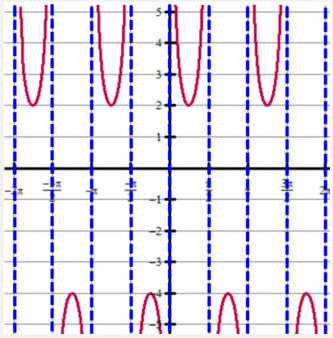
18. $\frac{\sqrt{3}}{3}$

19. 1.057

20. -1.236

21. -0.228

22.

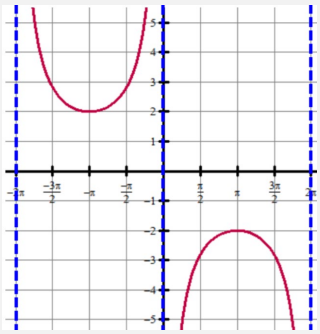


Range: $y\le -4$ and $y\ge 2$

Vertical Asymptotes:

$\frac{\pi}{2}n$ where n is an integer

23.



Range: $y\le -2$ and $y\ge 2$

Vertical Asymptotes:

$2\pi n$ where n is an integer

24. $\csc x$

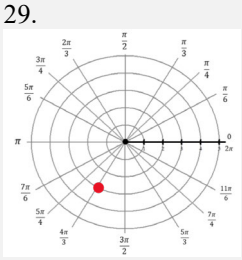
25. $\sin^2 x$

26. $\csc x$

27. $\frac{-\sqrt{6}-\sqrt{2}}{4}$

28. $\frac{\sqrt{3}}{2}$

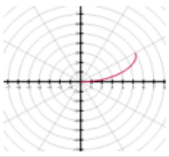
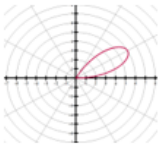
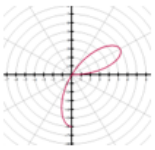
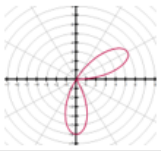
29.



Answers to Unit 3B Corrective Assignment (continued)

30. $\left(-\frac{5\sqrt{3}}{2}, \frac{5}{2}\right)$	31. $(2\sqrt{5},\ 4.248)$	32. $\sqrt{34}(\cos 2.111 + i\sin 2.111)$			33. $3\sqrt{2} + 3i\sqrt{2}$
34. C	35. E	36. I	37. F	38. A	39. L
40. J	41. H	42. B	43. G	44. D	45. K
46a. increasing	46b. increasing	46c. 3.445	46d. $y - 4.913 = 3.445\left(x - \frac{\pi}{8}\right)$ $f\left(\frac{\pi}{4}\right) \approx 6.265$		

47.

Graph of $r = f(x)$				
Interval	$0 \le x \le \frac{\pi}{6}$	$\frac{\pi}{6} \le x \le \frac{\pi}{3}$	$\frac{\pi}{3} \le x \le \frac{\pi}{2}$	$\frac{\pi}{2} \le x \le \frac{2\pi}{3}$
r is Pos/Neg	Positive	Positive	Negative	Negative
Distance from the origin Inc/Dec	Increasing	Decreasing	Increasing	Decreasing